

Bayesian statistics

Principle and applications

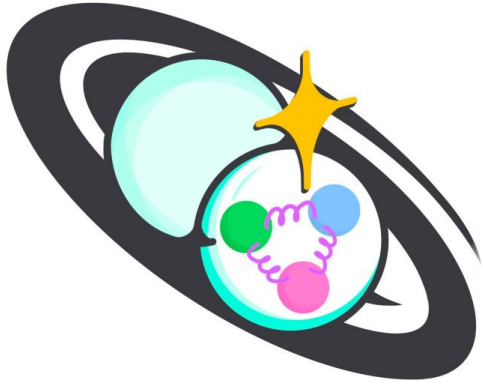
Thibaud Wouters

With contributions from Peter T. H. Pang

Outline

- What is NMMA and why does it need Bayesian statistics?
- Bayesian inference
 - Parameter estimation
 - Hypothesis testing
 - Computational methods
- A vision into the future of NMMA?

NMMA



NMMA

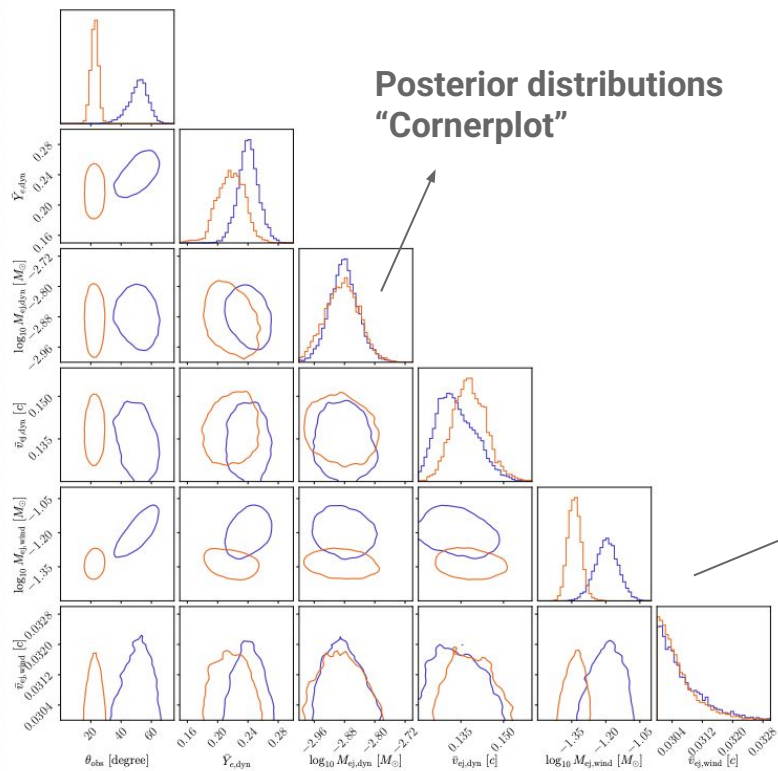
Nuclear Multi-Messenger Astronomy

What is NMMA?

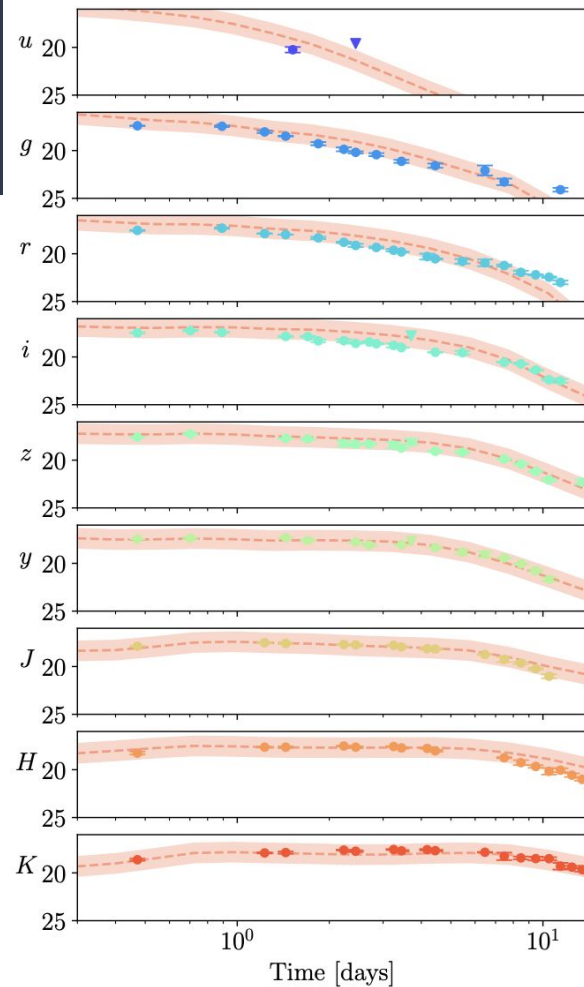
- A pythonic library for probing nuclear physics and cosmology with multimessenger analysis
 - Bayesian parameter estimation and hypothesis testing
 - Single or multi-component electromagnetic signals
 - Gravitational-wave signals
 - Multi-messenger events, i.e., EM+GW
 - Others
 - Build kilonova surrogate model
 - Calculate prior on EOS with given observations
 - Flexible sampler interface (using bilby)

AT2017gfo

Posterior distributions
“Cornerplot”



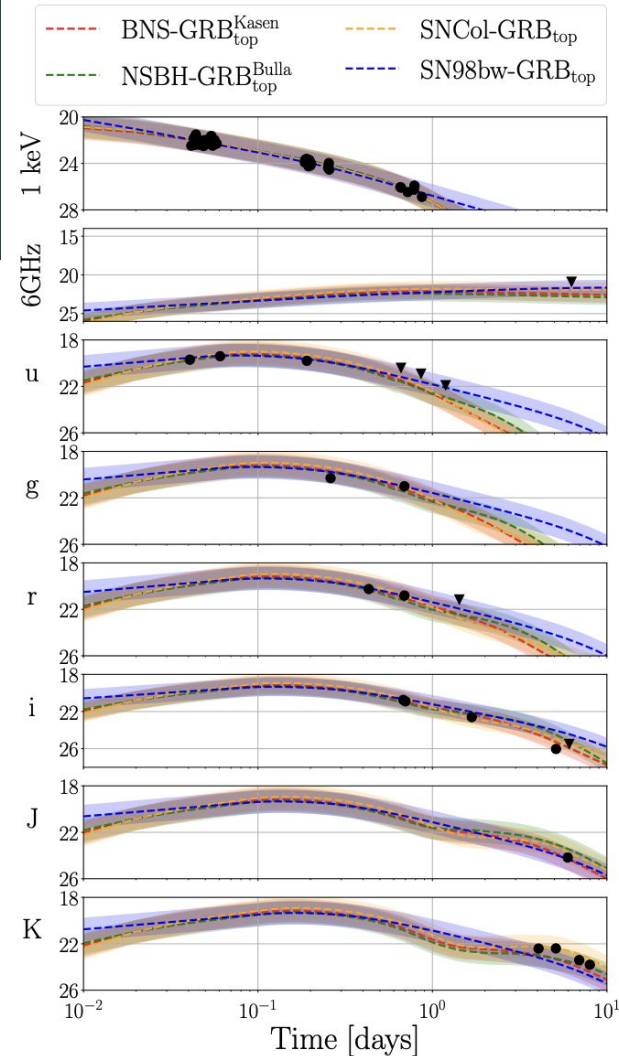
Posterior predictive
distributions



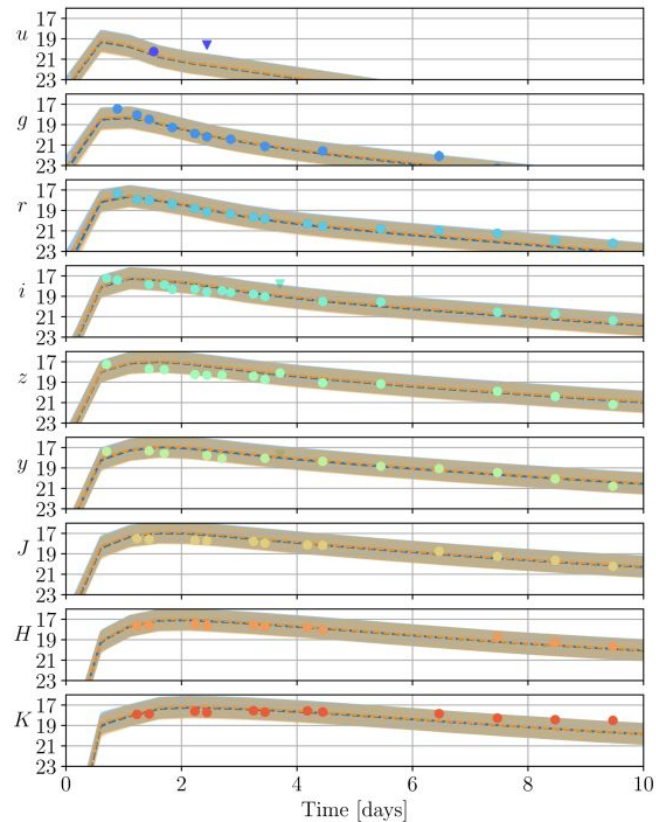
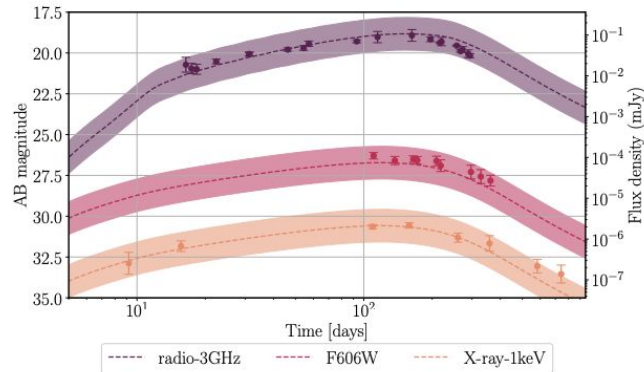
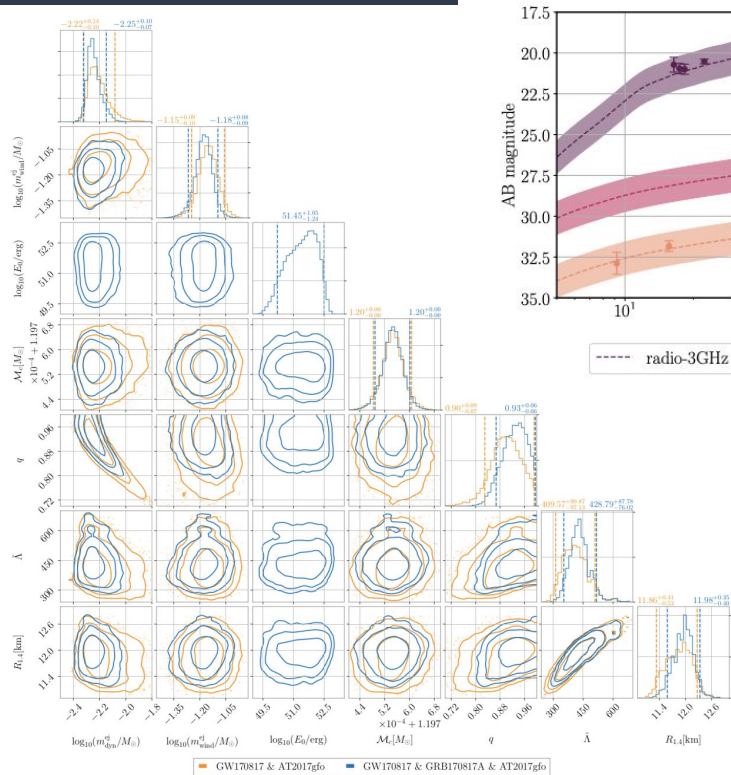
GRB211211A

Name	Astrophysical Processes	GRB Jet Structure	Model dimension	Bayes factor $\ln[\mathcal{B}_{\text{ref}}^1]$	Likelihood $\ln[\mathcal{L}_{\text{ref}}^1(\hat{\theta})]$
BNS-GRB _{top} ^{Kasen}	Kilonova + GRB	Tophat	11	ref.	ref.
BNS-GRB _{Gauss} ^{Kasen}	Kilonova + GRB	Gaussian	12	-1.01 ± 0.10	-0.33
BNS-GRB _{top} ^{Bulla}	Kilonova + GRB	Tophat	11	-0.49 ± 0.10	-1.15
BNS-GRB _{Gauss} ^{Bulla}	Kilonova + GRB	Gaussian	12	-1.59 ± 0.10	-2.13
NSBH-GRB _{top}	Kilonova + GRB	Tophat	11	-3.76 ± 0.10	-3.82
NSBH-GRB _{Gauss}	Kilonova + GRB	Gaussian	12	-2.08 ± 0.10	-4.16
SNCOL-GRB _{top}	r CCSNe + GRB	Tophat	14	-10.42 ± 0.11	-3.04
SNCOL-GRB _{Gauss}	r CCSNe + GRB	Gaussian	15	-10.74 ± 0.11	-3.58
SN98bw-GRB _{top}	CCSNe + GRB	Tophat	8	-6.93 ± 0.10	-8.14
SN98bw-GRB _{Gauss}	CCSNe + GRB	Gaussian	9	-8.05 ± 0.10	-8.13
GRB _{top}	GRB	Tophat	8	-6.04 ± 0.10	-7.10
GRB _{Gauss}	GRB	Gaussian	9	-6.96 ± 0.10	-7.33

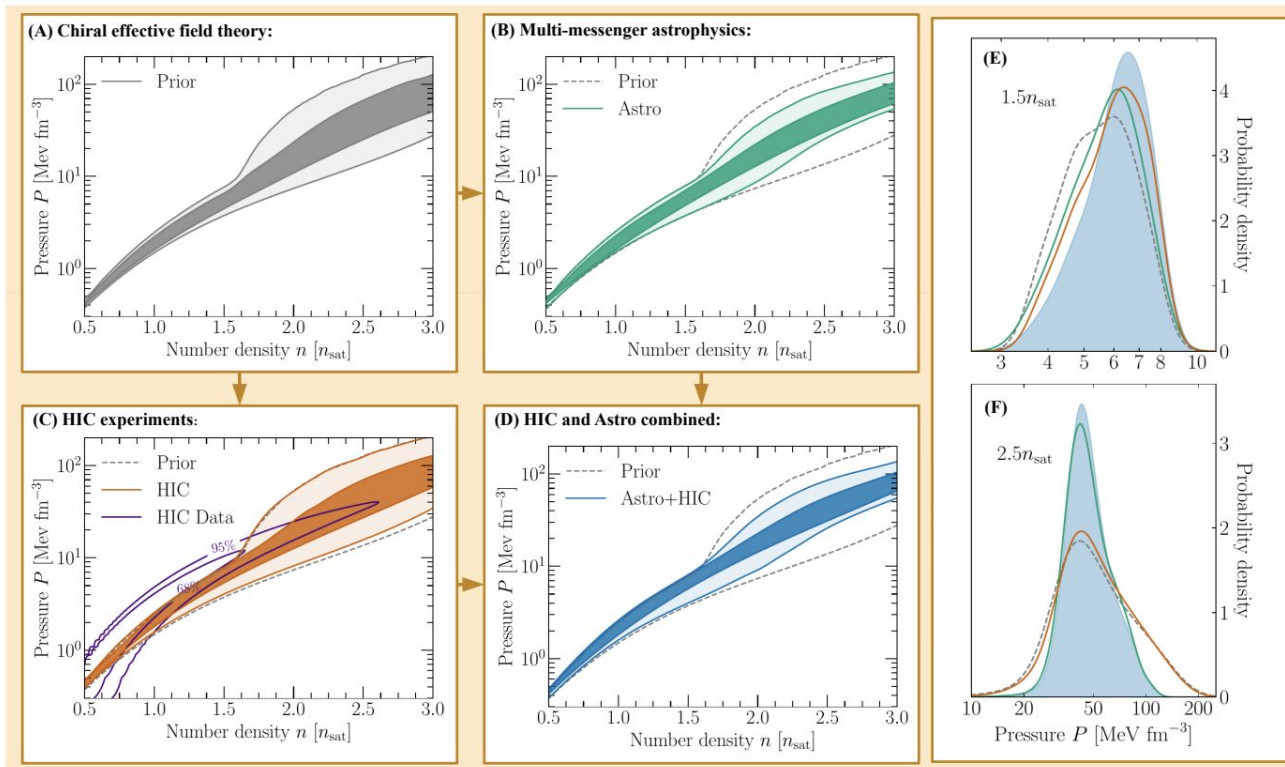
Preference for models



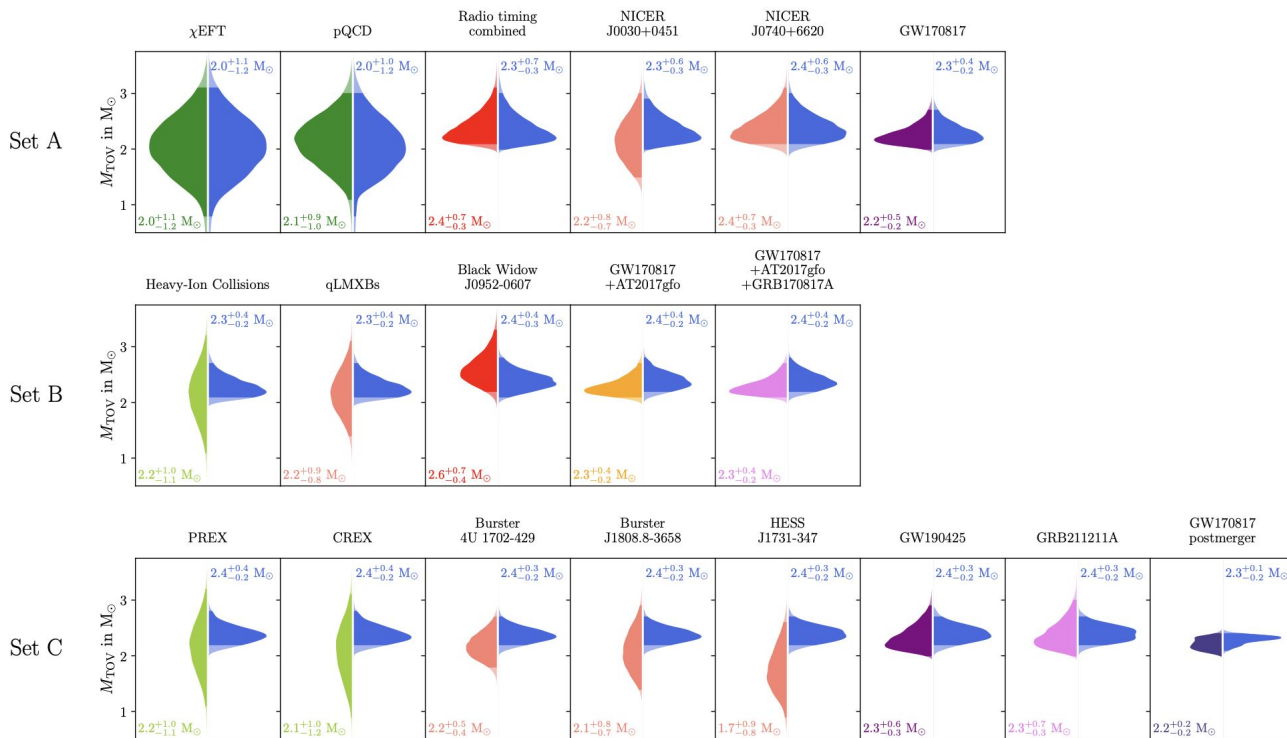
GW170817-AT2017gfo-GRB170817A



Astrophysics + Heavy ion collision

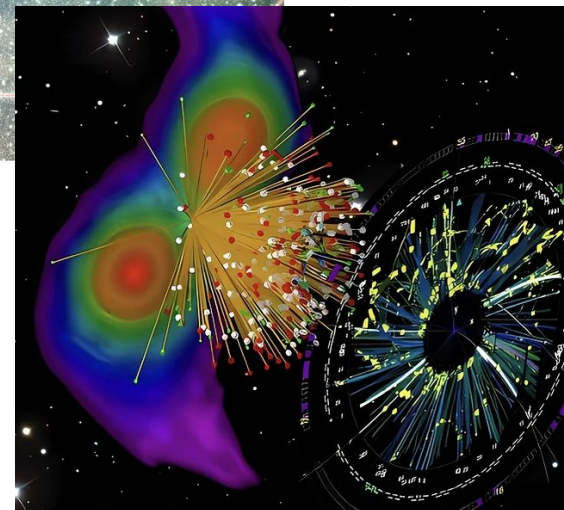
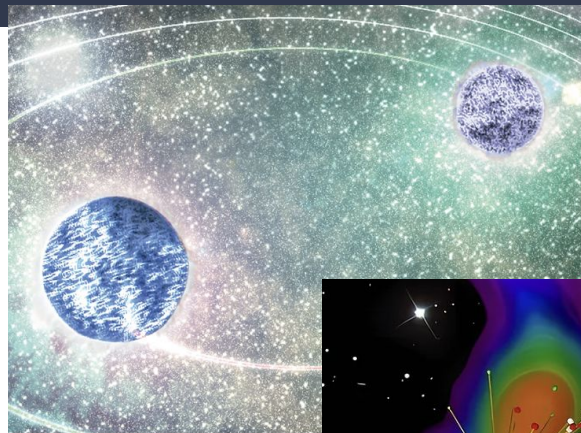


Combining various constraints (astro+nuclear)



Why Bayesian?

- Essential for answering physics question
 - Data analysis
 - Discoveries
- Unlock the secret from neutron stars
 - Multi-messenger astronomy
 - Supranuclear matter
 - Hubble constant
 - Nucleosynthesis
 - More...



What is a Bayesian inference problem?

- **Solve hard questions:** Given the observed data, how likely is a hypothesis true?
 - Given the results of 100 dice rolls, how likely the dice is fair?
 - Given the AT2017gfo, what is the ejecta mass?
 - **The posterior**
- **With an easier question:** Under a hypothesis, what will the observed data be?
 - Given a dice is fair, what is the probability of getting a “1”?
 - Given a kilonova model and ejecta mass, what are the resulting lightcurves?
 - **The likelihood**
- *How do we connect the two?*

Bayes theorem: parameter estimation

- The core of Bayesian inference is the **Bayes theorem**
- Tells us how to update **prior** beliefs (before observations) with data **likelihood** to get our **posterior** beliefs (after observations)

$$p(\vec{\theta}|d, \mathcal{H}) = \frac{p(d|\vec{\theta}, \mathcal{H})p(\vec{\theta}|\mathcal{H})}{p(d|\mathcal{H})}$$

$$\mathcal{P}(\vec{\theta}) = \frac{\mathcal{L}(\vec{\theta})\pi(\vec{\theta})}{Z}$$

$\mathcal{P}(\vec{\theta})$: Posterior

$\mathcal{L}(\vec{\theta})$: Likelihood

$\pi(\vec{\theta})$: Prior

Z : Evidence

Bayes theorem: parameter estimation

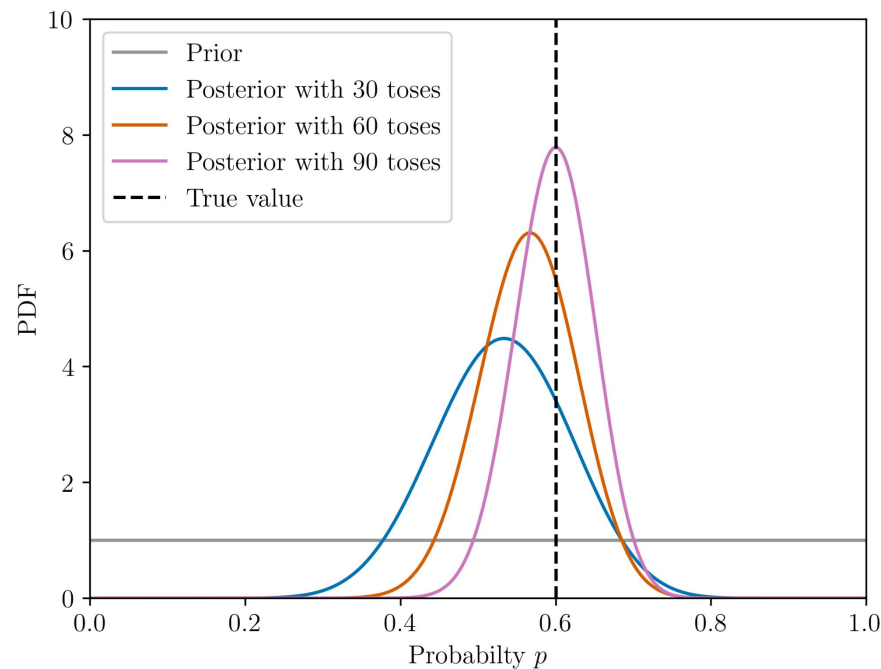
- Bayesian probability is always conditioned
- Henrik does not like superhero movies
 - What is it conditioned on? Is it a prior or posterior?
- Henrik has not seen superhero movies in years
 - Prior to experiments
 - Prior
- Henrik regularly watch superhero movies
 - After the experiments
 - Posterior



Bayes theorem: parameter estimation

- What is the probability of getting a tail from a coin given a series of n tosses
- Prior
 - Uniform prior (“agnostic”)
- Likelihood
 - Binomial distribution

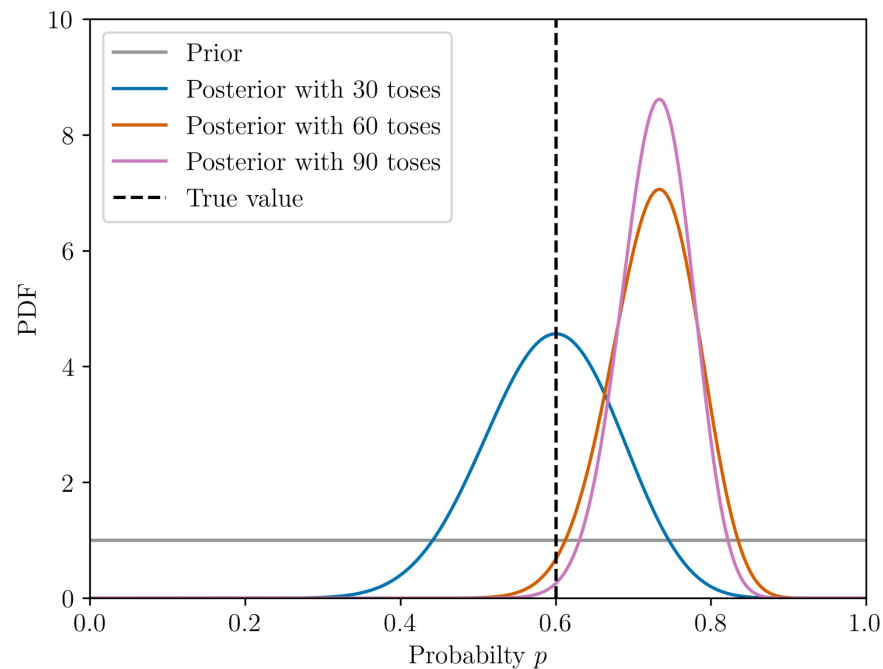
$$\text{Prob}(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$



Bayes theorem: parameter estimation

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$$\text{Prob}(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$



Bayes theorem: hypothesis testing

- Same Bayes rule, but used for quantifying how likely a hypothesis is true
- If equal prior odds (often): Bayes factor indicate preference for a hypothesis
 - Ratio of evidences $\rightarrow$ log BF is difference of evidences
 - Simpler models preferred (“Occam’s razor”)

$$p(\mathcal{H}|d, I) = \frac{p(d|\mathcal{H}, I)p(\mathcal{H}|I)}{p(d|I)}$$

$$\frac{p(\mathcal{H}_1|d, I)}{p(\mathcal{H}_2|d, I)} = \frac{p(d|\mathcal{H}_1, I)p(\mathcal{H}_1|I)}{p(d|\mathcal{H}_2, I)p(\mathcal{H}_2|I)}$$

$$\mathcal{O}_2^1 = \mathcal{B}_2^1 \Pi_2^1$$

$\mathcal{O}_2^1$:Odds ratio

$\mathcal{B}_2^1$:Bayes factor

Π_2^1 :Prior odds

Bayes theorem: hypothesis testing

- **Occam's razor:** If two hypotheses explain the data equally well, the simpler model should be more plausible.

$\mathcal{H}_1 :$

$$p(d|\mathcal{H}_1, I) = \mathcal{L}_0$$

$\mathcal{H}_2 :$

$$p(d|\theta, \mathcal{H}_2, I) = \mathcal{L}_0 \exp \left(\frac{1}{2} \left(\frac{\theta - \theta_0}{\sigma} \right)^2 \right)$$

$$p(\theta|\mathcal{H}_2, I) \propto \exp \left(\frac{1}{2} \left(\frac{\theta - \theta_0}{\sigma_\theta} \right)^2 \right)$$

Bayes theorem: hypothesis testing

$\mathcal{H}_1 :$

$$p(d|\mathcal{H}_1, I) = \mathcal{L}_0$$

$\mathcal{H}_2 :$

$$\begin{aligned} p(d|\mathcal{H}_2, I) &= \int d\theta p(d|\mathcal{H}_2, I)p(\theta|\mathcal{H}_2, I) \\ &= \mathcal{L}_0 \sqrt{\frac{\sigma^2}{\sigma^2 + \sigma_\theta^2}} \end{aligned}$$

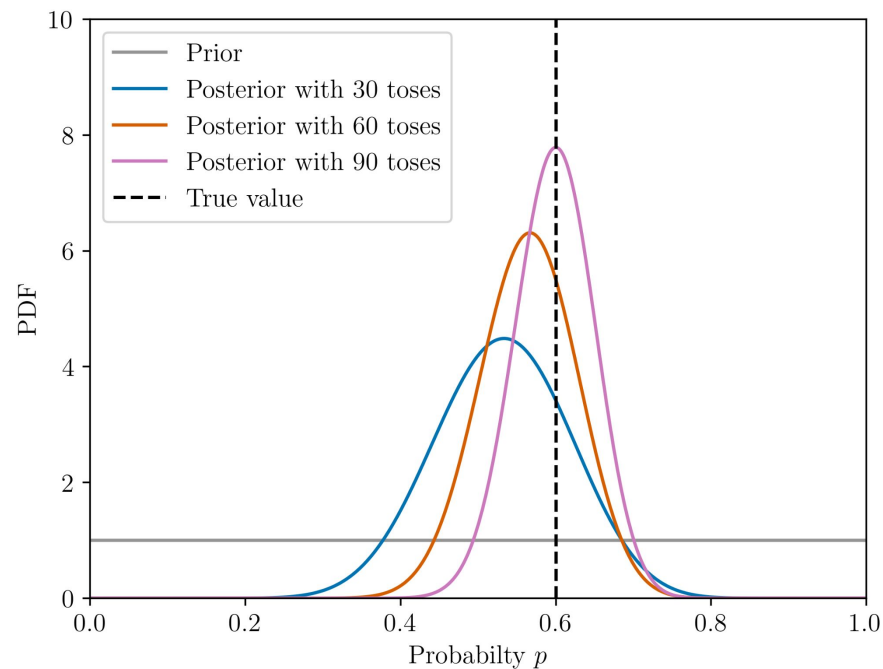
Bayes theorem: hypothesis testing

$$\mathcal{O}_2^1 = \Pi_2^1 \sqrt{1 + \frac{\sigma_\theta^2}{\sigma^2}}$$

- Hypothesis 2 is penalized due to its complexity:
 - Number of parameters
 - Allowed ranges of parameters (prior widths)

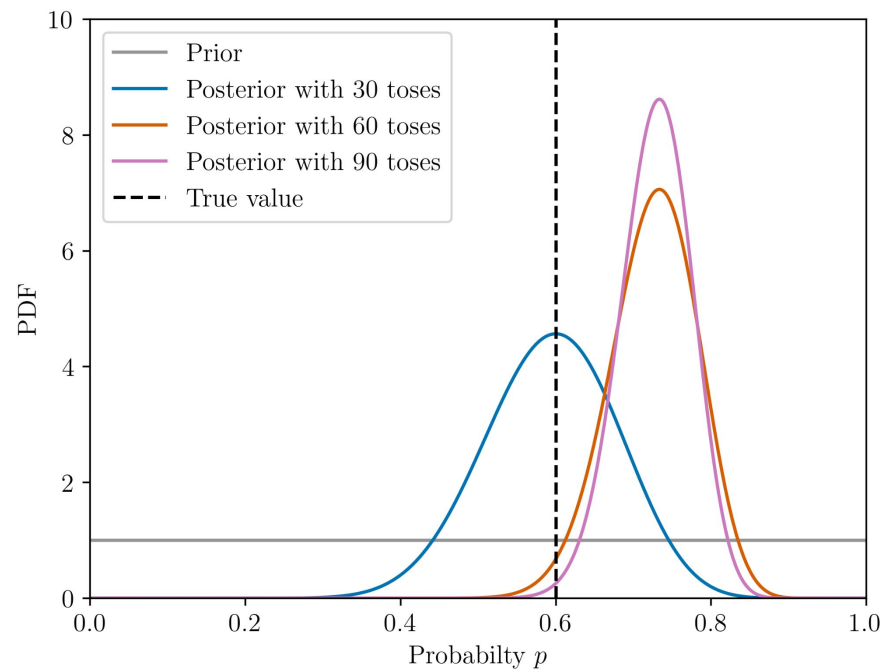
Bayes theorem: hypothesis testing

Amount of toss	Bayes factor of fair against unfair
30	4.198498
60	3.697612
90	1.271816



Bayes theorem: hypothesis testing

Amount of toss	Bayes factor of fair against unfair
30	2.497146
60	7.9156e-3
90	3.233e-4



Bayes theorem: hypothesis testing

- “What is a large Bayes factor?”
 - Jeffrey’s scale: heuristic
 - Background-foreground distributions (costly, therefore less common)

$\log_{10}(\mathcal{B}_2^1)$	Interpretation
$[0, \frac{1}{2}]$	Barely worth mentioning
$[\frac{1}{2}, 1]$	Substantial evidence
$[1, \frac{3}{2}]$	Strong evidence
$[\frac{3}{2}, 2]$	Very strong evidence
> 2	Decisive evidence

The slide of utter disappointment

- ***Posterior distributions and their normalization constants are hard to compute***
- Often, we can just only call the likelihood at arbitrary parameter point
- So in the following slides, we'll see and compare methods to
 - Sample from the posterior
 - Compute its normalization constant
 - Or both

Computational methods

- Estimating the posterior distribution and/or compute Bayesian evidence
 - **Monte Carlo integration and importance sampling**
 - Markov chain Monte Carlo (MCMC)
 - Parallel-tempering MCMC
 - Nested sampling (→ NMMA today)
 - Sequential Monte Carlo
- Future?

Monte Carlo

- Suppose posterior has no closed form, but we have N samples from it (“somehow”)
- **Monte Carlo** = repeated random sampling to estimate numerical results
 - Allows to compute expectations with the samples

$$p(\vec{\theta}|d, \mathcal{H}) = \frac{p(d|\vec{\theta}, \mathcal{H})p(\vec{\theta}|\mathcal{H})}{p(d|\mathcal{H})}$$

$$\mathbb{E}_P[f(\theta)] = \int f(\theta)P(\theta) \, \mathrm{d}\theta \approx \frac{1}{N} \sum_{i=1}^N f(\theta_i)$$

Crux of sampling

- Big question: How do we obtain samples from the posterior? Sampling techniques!
- The simplest: **importance sampling**
 - Use “cheap” distribution as generator, correct for target distribution
 - Proposal density Q

$$\int f(\theta)P(\theta) \, d\theta = \int f(\theta) \frac{P(\theta)Q(\theta)}{Q(\theta)} Q(\theta) \, d\theta \qquad \mathbb{E}_P[f(\theta)] = \mathbb{E}_Q \left[\frac{f(\theta)P(\theta)}{Q(\theta)} \right]$$

- i.i.d samples from Q get **importance weight** $w_i = \frac{P(\theta_i)}{Q(\theta_i)}$

Computational methods

- Estimating the posterior distribution and/or compute Bayesian evidence
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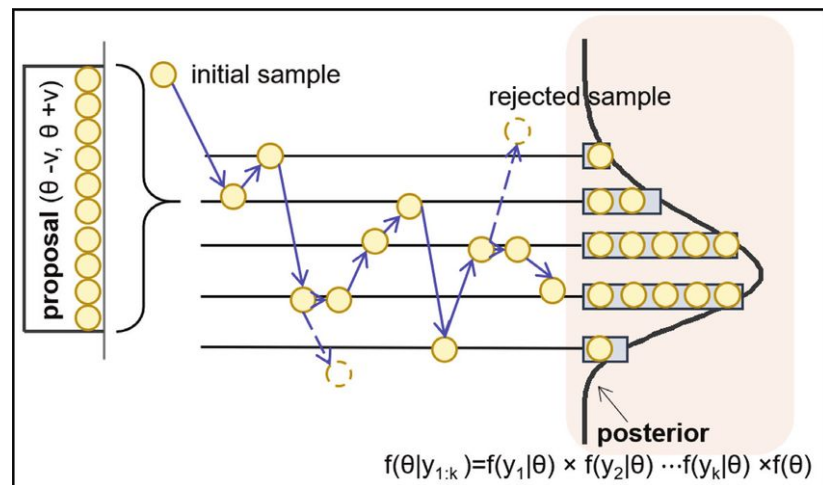
Markov chain Monte Carlo

- Collection of Markov chains:
 - One chain: $X_0, X_1, X_2, \dots, X_t$ with $P(X_{n+1}|X_1, \dots, X_n) = P(X_{n+1}|X_n)$
- Metropolis-Hastings algorithm:

- Proposal for new X_{t+1}
- Accept-reject step

$$a = \min \left(1, \frac{\bar{p}(\theta')}{\bar{p}(\theta_t)} \frac{Q(\theta_t|\theta')}{Q(\theta'|\theta_t)} \right)$$

- Reject? $X_{t+1} = X_t$



Markov chain Monte Carlo – Kernels

$$P(X_{n+1}|X_n)$$

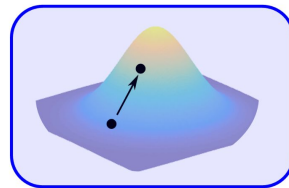
How do we propose a new position? A few options

- **Gaussian random walk**, from current position: cheap
- **Differential evolution**: difference 2 chains as jump (“global”)

$$\theta_{i+1}^j = \theta_i^j + \gamma \left[\theta_i^k - \theta_i^l + \epsilon \right], \quad \epsilon \sim \mathcal{N}(0, C_\epsilon)$$

- Gradient-based proposals, use local information likelihood (**MALA**)

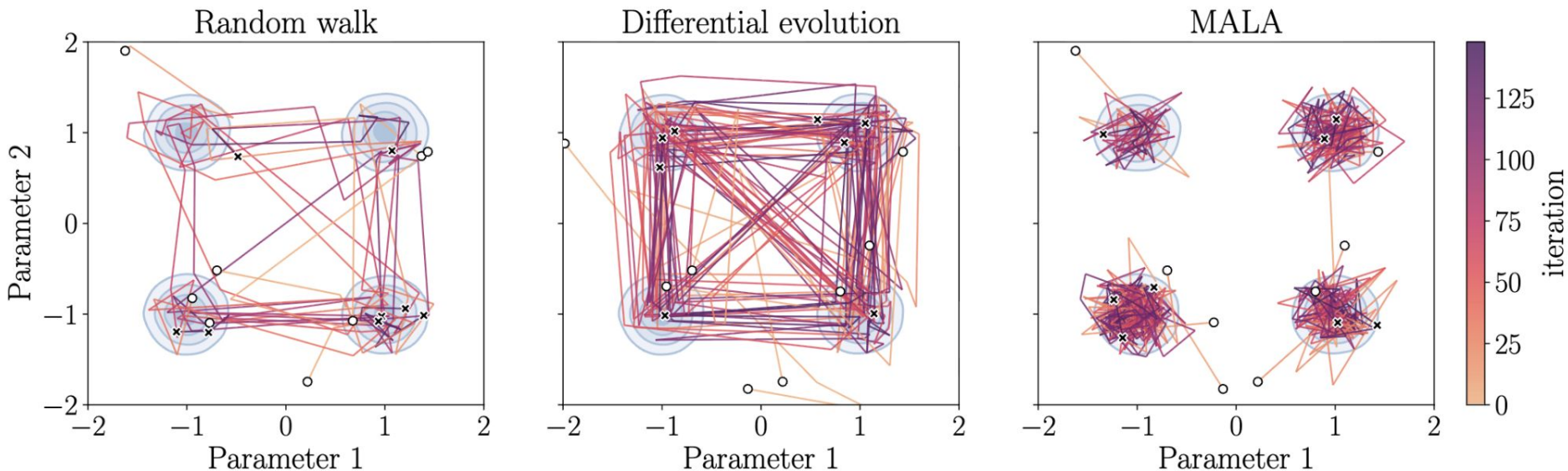
$$\theta_{i+1} \sim \mathcal{N} \left(\theta_i + \frac{1}{2} M \nabla \log \mathcal{P}(\theta_i), M \right)$$



- ... Many more!
- Note: Such kernels are also used in nested sampling, sequential Monte Carlo

Markov chain Monte Carlo – Kernels

$$P(X_{n+1}|X_n)$$



Markov chain Monte Carlo – (dis)advantages

- MCMC has a few issues
 - No clear **convergence criterion**
 - Does not compute evidence (Monte Carlo estimate unstable)
 - Only reach the posterior after a “**burn-in**”: discard ~50% of iterations
 - Positions are correlated: **thinning**
- Benefits:
 - Highly parallelizable: N chains run fully independent (except maybe for proposal tuning)

Markov chain Monte Carlo

- Given a transition probability $T(\theta', \theta)$; the probability distribution iterates as:

$$p_{t+1}(\theta') = \int d\theta T(\theta', \theta) p_t(\theta)$$

- Invariant / detailed balance w.r.t. the desired distribution $\bar{p}(\theta)$

$$\bar{p}(\theta') = \int d\theta T(\theta', \theta) \bar{p}(\theta) \quad T(\theta, \theta') \bar{p}(\theta') = T(\theta', \theta) \bar{p}(\theta)$$

- Ergodicity:

$$\lim_{t \rightarrow \infty} p_t(\theta) = \bar{p}(\theta), \text{ for any } p_0(\theta)$$

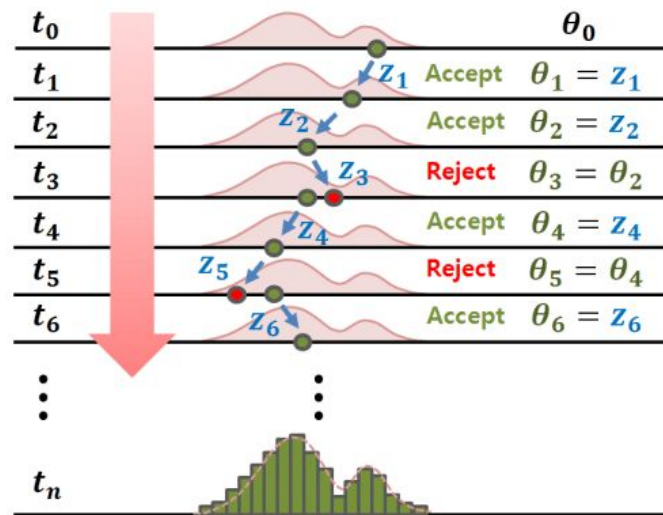
Metropolis-Hasting algorithm

- Given a current state θ_t , a new state θ' is proposed by $Q(\theta'|\theta)$
- The new state is accepted with a probability:

$$a = \min \left(1, \frac{\bar{p}(\theta')}{\bar{p}(\theta_t)} \frac{Q(\theta_t|\theta')}{Q(\theta'|\theta_t)} \right)$$

- If rejected, the state stays unchanged

$$\theta_{t+1} = \theta_t$$



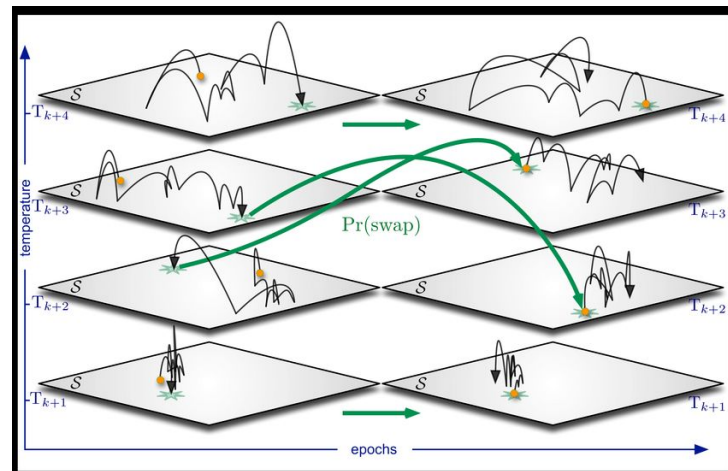
Parallel-tempering MCMC

- Failed to handle multimodal distribution and no evidence estimation

$$p_T(\theta|d) \propto p(d|\theta)^{1/T} p(\theta)$$

- With higher temperature
 - the smoother the target distribution
 - Less likely to be struck
- Swap locations between chains

$$r_s = \min \left(1, \left(\frac{p(d|\theta_j)}{p(d|\theta_i)} \right)^{1/T_i - 1/T_j} \right)$$



Parallel-tempering MCMC

- Thermodynamic integration

$$p(d|\mathcal{H}, \beta = 1/T) = \int d\theta p(d|\theta, \mathcal{H})^\beta p(\theta|\mathcal{H})$$

- Average over the posterior distribution at different temperature

$$\frac{d}{d\beta} \ln p(d|\mathcal{H}, \beta) = \langle \ln p(d|\theta, \mathcal{H}) \rangle_\beta$$

$$\begin{aligned} \ln p(d|\mathcal{H}) &= \ln p(d|\mathcal{H}, \beta = 1) \\ &= \int_0^1 d\beta \frac{d}{d\beta} \ln p(d|\mathcal{H}) + \ln p(d|\mathcal{H}, \beta = 0) \\ &= \int_0^1 d\beta \langle \ln p(d|\theta, \mathcal{H}) \rangle_\beta \end{aligned}$$

Computational methods

- Estimating the posterior distribution and/or compute Bayesian evidence
 - Monte Carlo integration and importance sampling
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 - Parallel-tempering MCMC
 - **Nested sampling (→ NMMA today)**
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Nested sampling – Idea

- Main goal is to estimate the **evidence**
- Getting posterior samples is a by-product

$$\mathcal{P}(\theta) = \frac{\mathcal{L}(\theta)\pi(\theta)}{Z}$$
$$Z = \int d\theta \mathcal{L}(\theta)\pi(\theta)$$

Nested sampling – Idea

- Convert the high-dimensional integration to one-dimensional integration

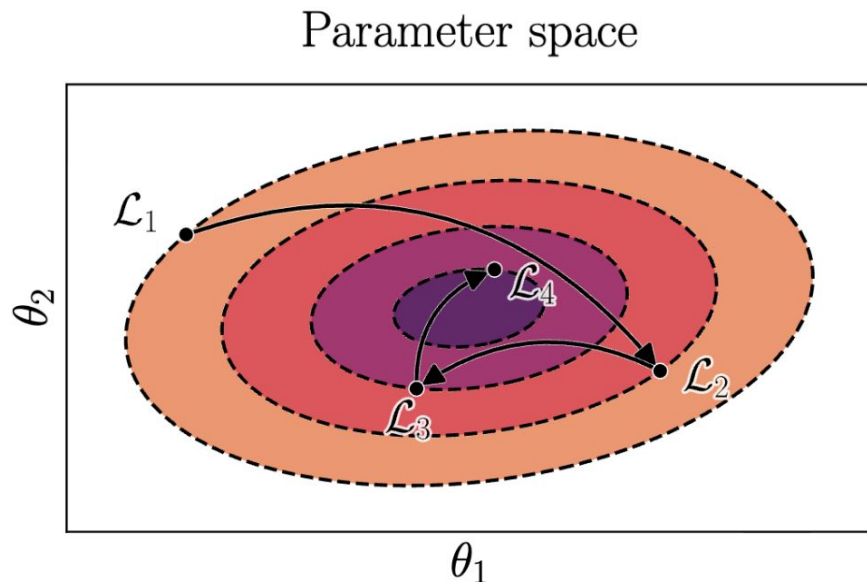
$$Z = \int d\theta \mathcal{L}(\theta) \pi(\theta) \rightarrow \int_0^1 dX \hat{\mathcal{L}}(X)$$

$$X(\lambda) = \int_{\theta: \mathcal{L}(\theta) > \lambda} d\theta \pi(\theta)$$

- X is the so-called **prior mass**
- $X = 0 \rightarrow$ hyper-volume enclosed by the *maximum* likelihood
- $X = 1 \rightarrow$ hyper-volume enclosed by the *minimum* likelihood

Nested sampling – Algorithm

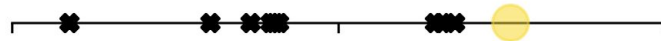
Live points explore parameter space, reaching higher and higher iso-likelihood contours



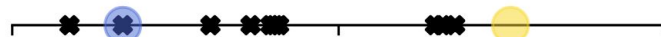
Uniformly distributed live points



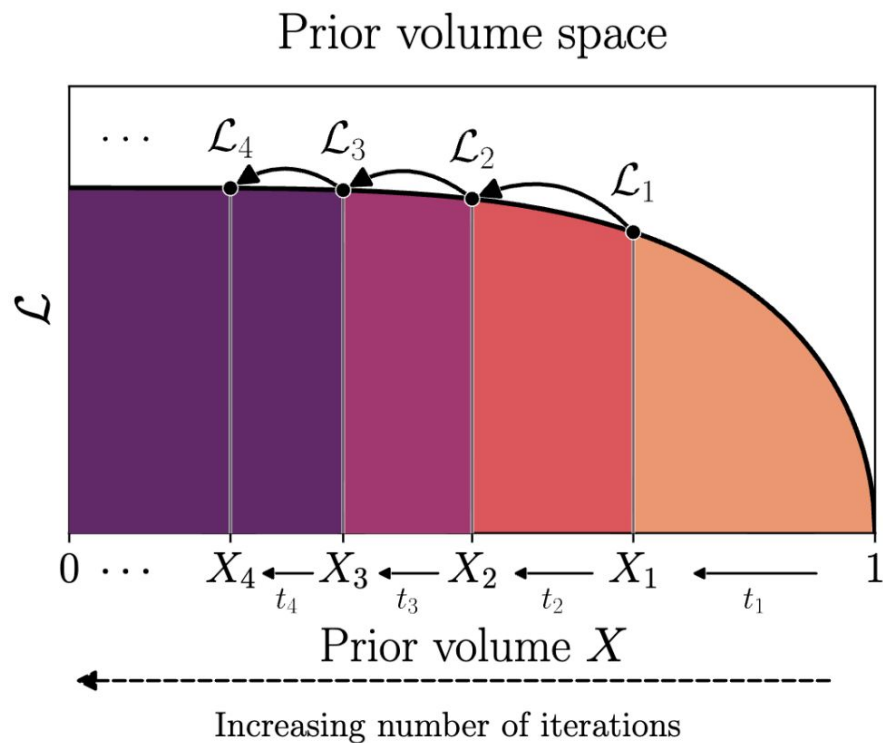
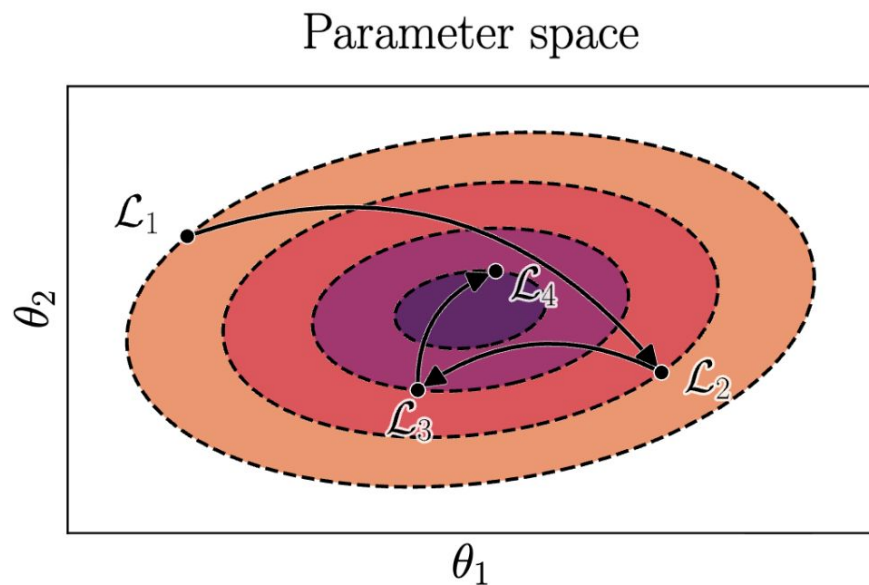
Remove worst



Draw replacement

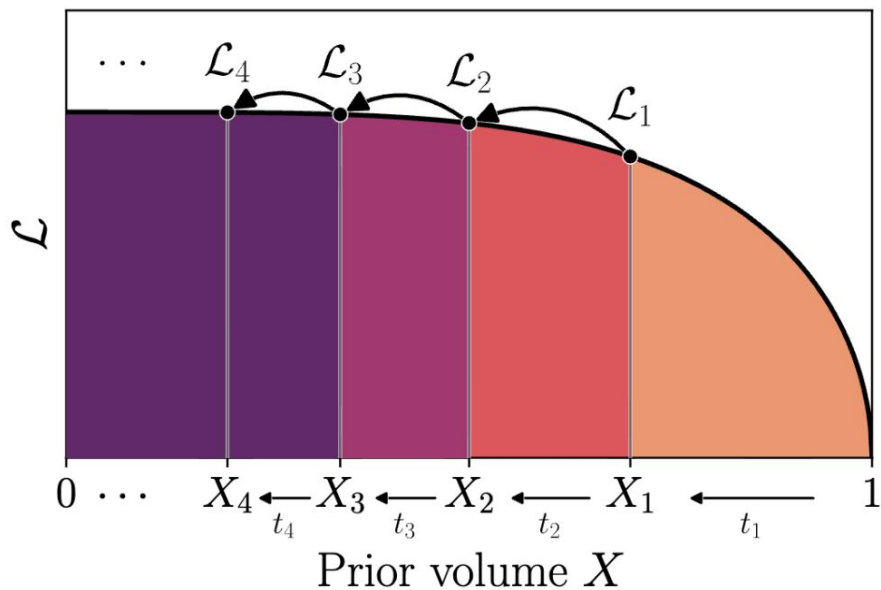


Nested sampling – Algorithm



Nested sampling – Algorithm

Prior volume space



The evidence is then simply the area under this curve $\rightarrow$ use Riemann integration on the set of (X_i, L_i) so-called “**dead points**”

Nested sampling – Evidence estimate

- The evidence is then computed by simple Riemann integration

$$Z = \int_0^1 dX \hat{\mathcal{L}}(X) \approx \frac{1}{2} \sum_{i=1} (\mathcal{L}_{i-1} + \mathcal{L}_i) (X_{i-1} - X_i) \equiv \sum_{i=1} w_i.$$

- But wait how do we compute $X(\lambda) = \int_{\theta: \mathcal{L}(\theta) > \lambda} d\theta \pi(\theta)$ at each iteration?
- **We don't!** We rely on the statistics to get educated guesses, and we move on with life

$$\ln X_{k+1} = \ln X_k + \ln t_k \quad t \sim \text{Beta}(n_{\text{live}}, 1)$$

Probabilistic prior mass

- Prior mass:

$$X(\lambda) = \int_{\theta: \mathcal{L}(\theta) > \lambda} d\theta \pi(\theta)$$

- If samples are following the prior distribution
- The prior mass follows a uniform distribution between 0 and 1 (probability integral transform)
- Therefore, the cumulative distribution function of X is given by

$$F(X) = \int_0^X dX' = X$$

Probabilistic prior mass

- Given K prior masses, the probability for a random variable χ to be higher than all of them is given by

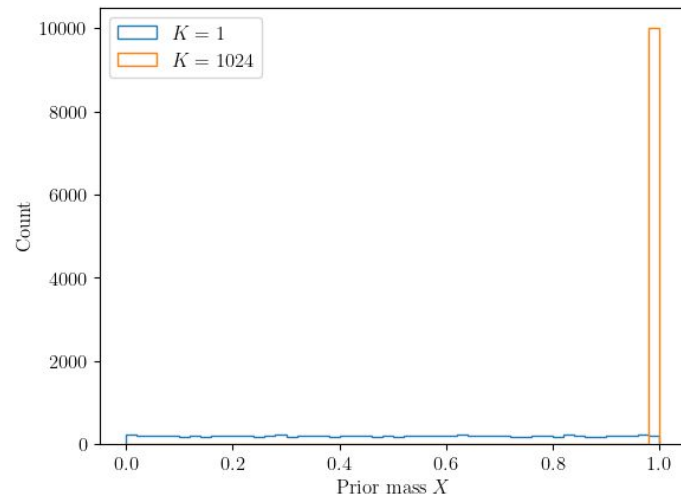
$$\Pr(\{X_i\} < \chi) = \prod_{i=0}^{N-1} F(X_i = \chi) = \chi^K$$

- The corresponding probability density is

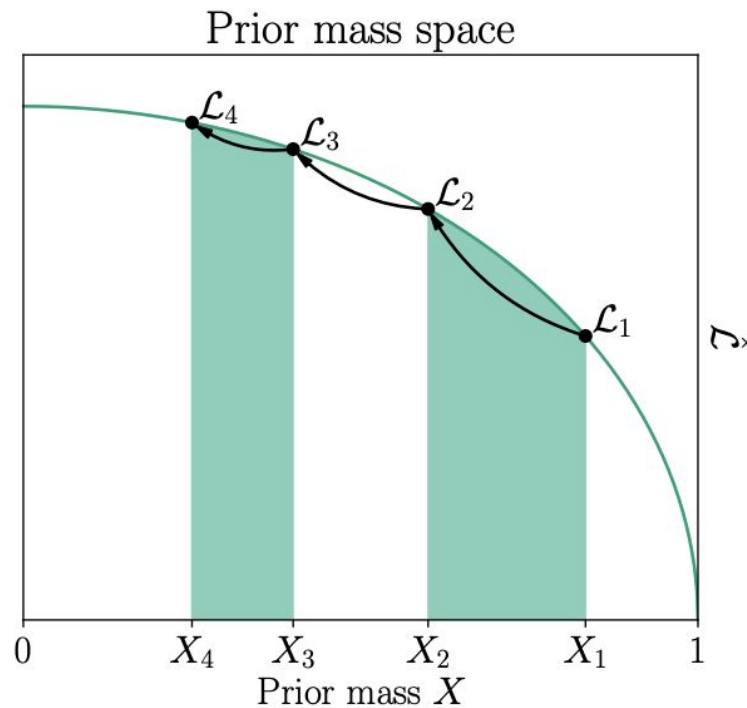
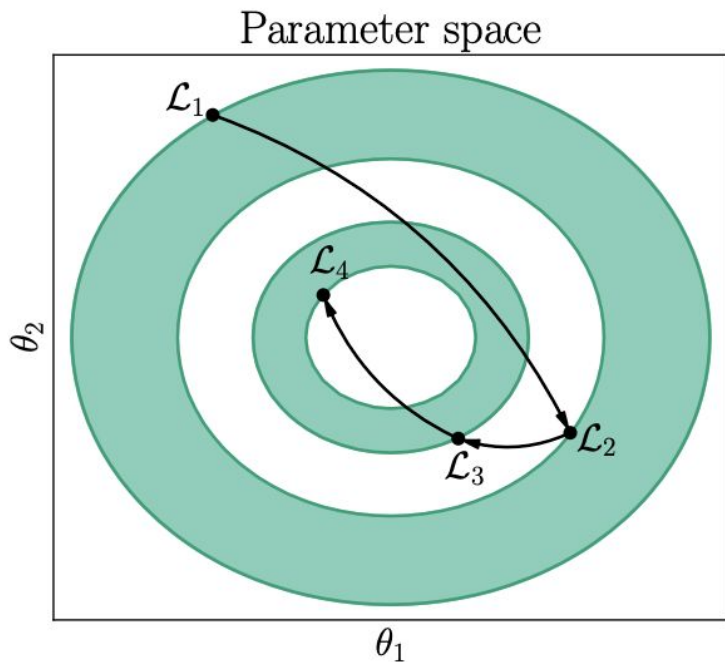
$$p(\chi) = \frac{d}{d\chi} \Pr(\{X_i\} < \chi) = K\chi^{K-1}$$

- If the prior masses and χ are bounded by some value X^*

$$p(t = \chi/X^*) = Kt^{K-1}, \text{ where } t \in [0, 1]$$

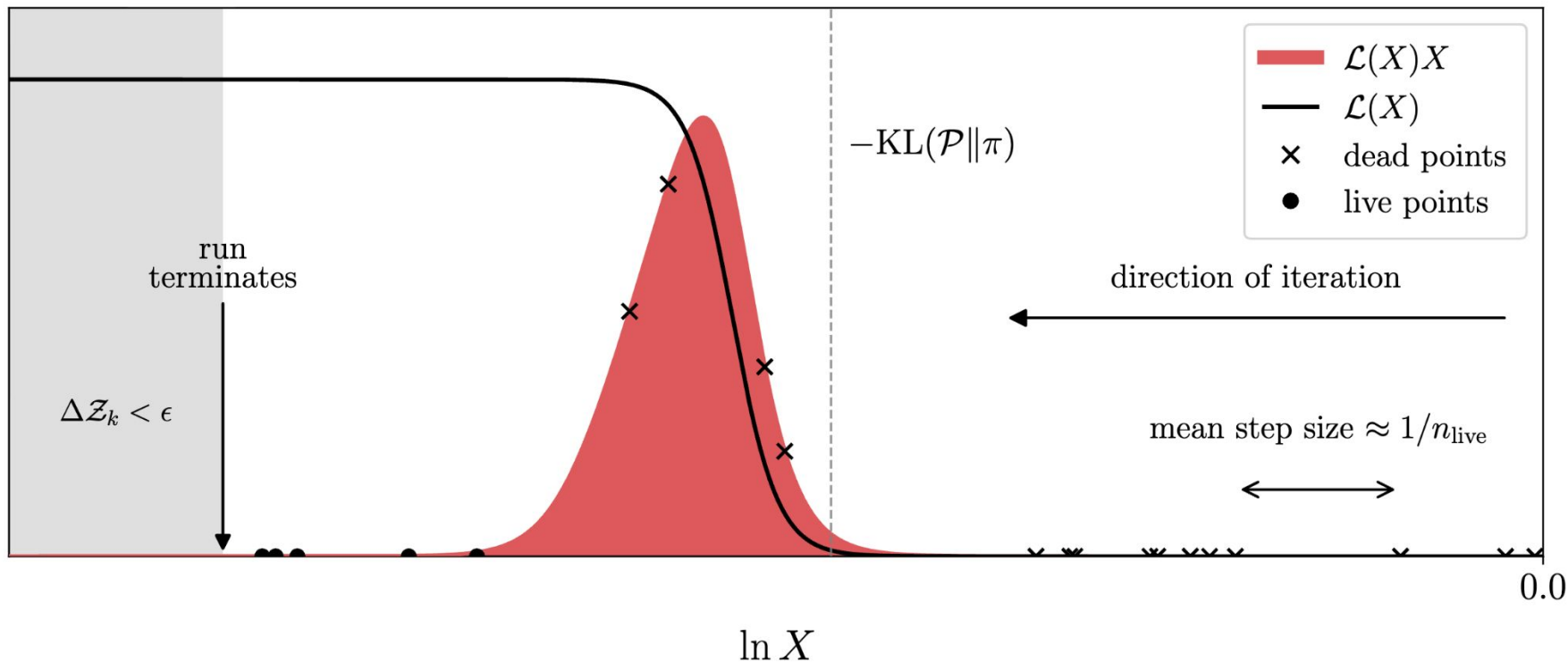


Nested sampling



Nested sampling – Progression

$$\mathcal{Z} = \int_{-\infty}^0 \mathcal{L}(X) X \, d \ln X$$



Nested sampling – Practical tips

- Time to reach the bulk ($\sim$ to convergence)

$$n_{\text{iter}} > n_{\text{bulk}} = \text{KL}(\mathcal{P}||\pi) \times n_{\text{live}}$$

- Uncertainty in evidence estimate

$$\Delta \ln \mathcal{Z} \approx \sqrt{\text{KL}(\mathcal{P}||\pi)/n_{\text{live}}}$$

- So best practices:
 - n_{live} should balance evidence estimate (larger) and runtime (lower)
 - Priors matter for runtime as well: balance between agnostic and informed

Nested sampling – Posterior samples

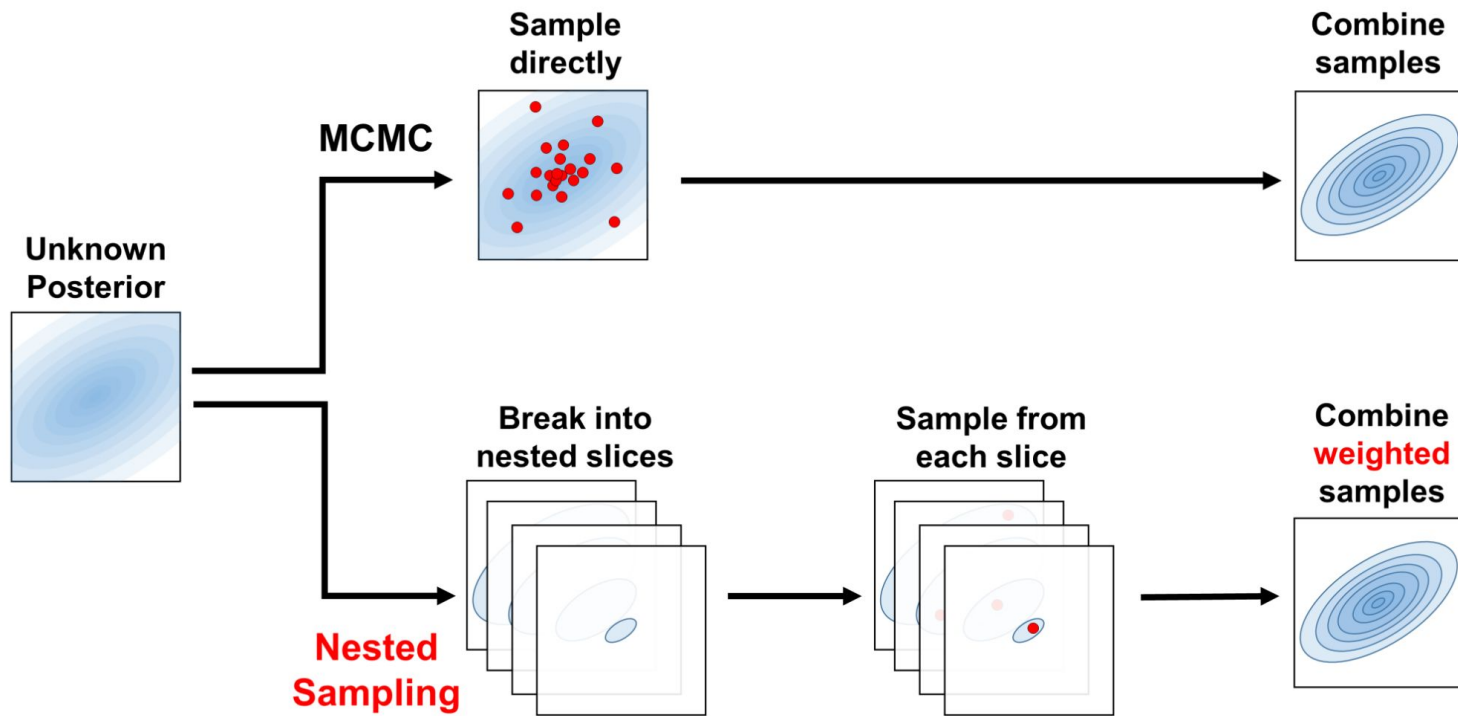
- The evidence is then computed by simple Riemann integration

$$Z = \int_0^1 dX \hat{\mathcal{L}}(X) \approx \frac{1}{2} \sum_{i=1} (\mathcal{L}_{i-1} + \mathcal{L}_i)(X_{i-1} - X_i) \equiv \sum_{i=1} w_i.$$

- The “dead points” are resampled into posterior samples using the **weights**

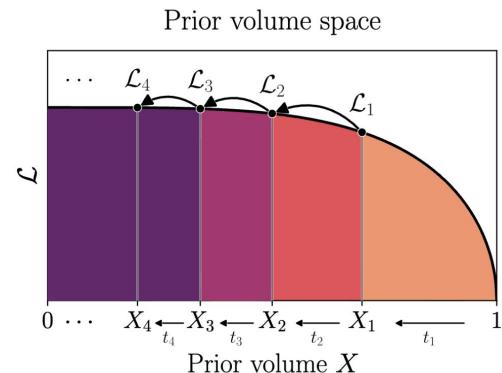
$$p_i = \frac{w_i}{\sum_{i=1} w_i}$$

Nested sampling – Comparison with MCMC

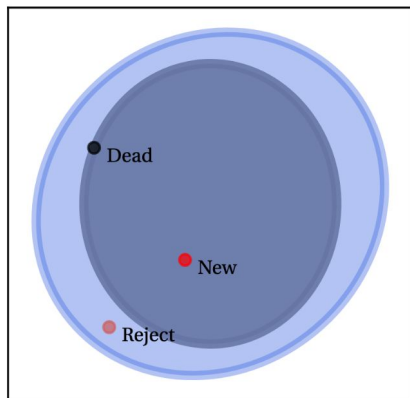


Nested sampling – Implementations

- Sampling from the likelihood constrained prior
 - **Propose points**, accept if $L > L^*$ (X smaller)
- Region samplers: $D < 10$, MultiNest
- Step samplers: $10 < D < 20$: dynesty (= bilby)

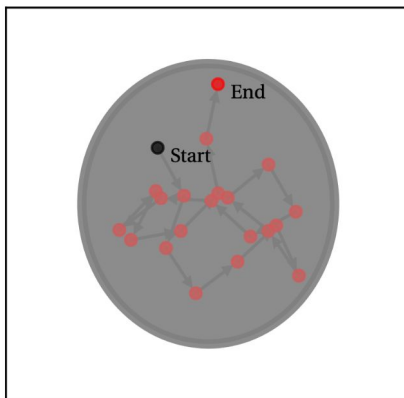


Region sampling

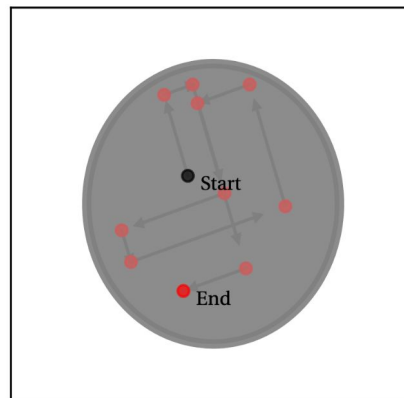


Ellipsoid sampling

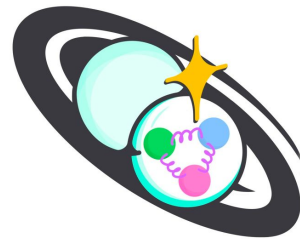
Step sampling



Random walk

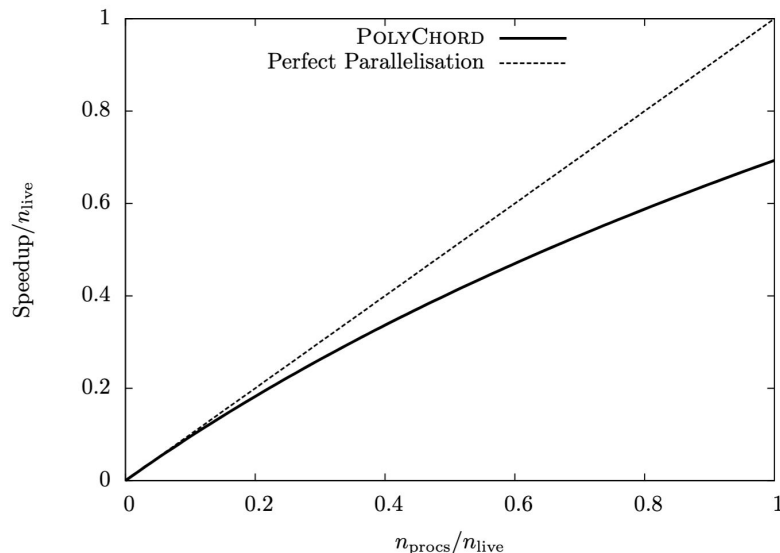


Slice sampling



Nested sampling – (dis)advantages

- Nested sampling is popular due to
 - Evidence estimate
 - Clear convergence criterion
- But has a few problems:
 - Speedup from parallelization is limited
 - Less flexible (always restart from prior)
 - But some [recent progress](#)



Computational methods

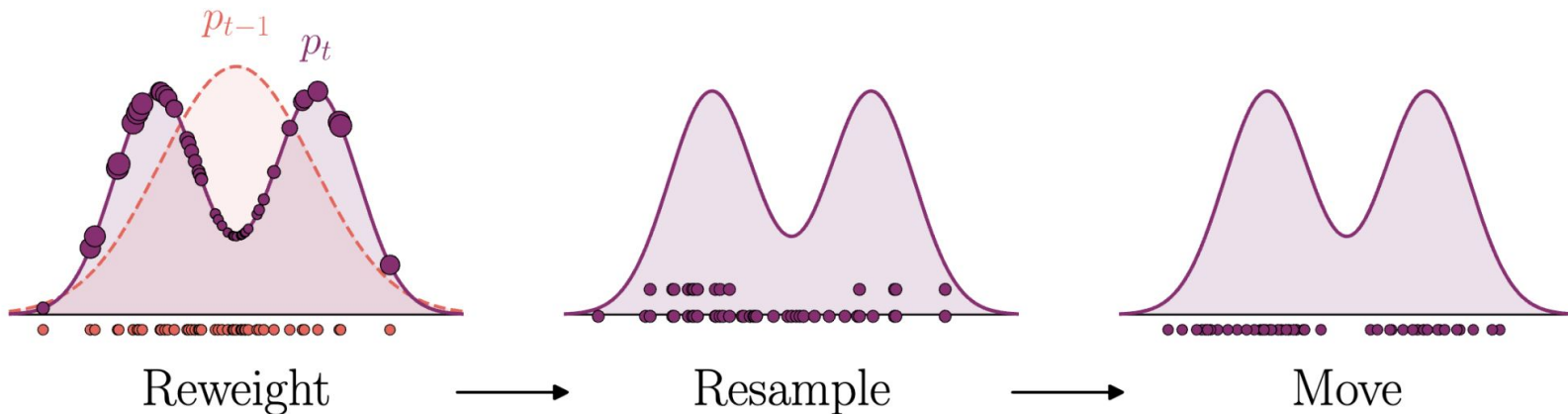
- Estimating the posterior distribution and/or compute Bayesian evidence
 - Monte Carlo integration and importance sampling
 - Markov chain Monte Carlo (MCMC)
 - Parallel-tempering MCMC
 - Nested sampling (→ NMMA today)
 - **Sequential Monte Carlo**
- Future?

Sequential Monte Carlo – an ideal combo

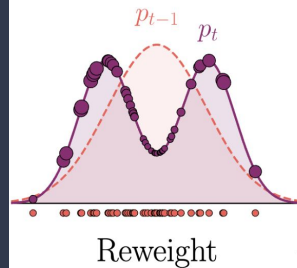
- Nested sampling
 - Computes the evidence
 - Has a clear convergence criterion
- MCMC
 - Is easy to parallelize
- Is there something that combines the two?
 - **YES:** Sequential Monte Carlo

Sequential Monte Carlo

- Set of N “particles” distributed according to p_t
- Bridge from prior to posterior using intermediate distributions p_t , $t = 0, \dots, T$
- One “bridge step” to go from p_t to p_{t+1} : Reweight $\rightarrow$ Resample $\rightarrow$ Move



Sequential Monte Carlo – Reweight



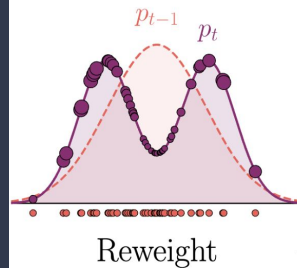
- Use p_t as a proposal distribution for p_{t+1} (current particles are proposals)
- Compute importance weights

$$w_t^i = \frac{\gamma_t(\theta_{t-1}^i)}{\gamma_{t-1}(\theta_{t-1}^i)} \quad p_t = \gamma_t / \mathcal{Z}_t$$

- Recursively tracks the evidence (prior is normalized)

$$\frac{\mathcal{Z}_t}{\mathcal{Z}_{t-1}} = \frac{1}{N} \sum_{i=1}^N w_t^i$$

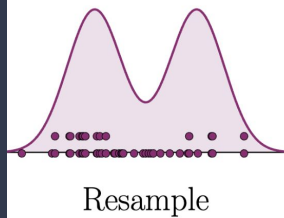
Sequential Monte Carlo – Reweight



- Due to the reweighting, we “lose” diversity in the N samples
- **Effective sample size (ESS)**: the weighted sample has similar estimate as ESS number of iid samples drawn from the target density
- $1 < \text{ESS} < N$
- Bad proposal density $\rightarrow$ low ESS $\rightarrow$ biased posteriors

$$\text{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}$$

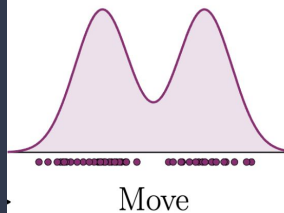
Sequential Monte Carlo – Resample



- Have N particles with weights
- Normalize weights
- Resample according to weights

$$W_t^i = \frac{p_t(\theta_{t-1}^i)}{p_{t-1}(\theta_{t-1}^i)} = \frac{Z_{t-1}}{Z_t} w_t^i$$

Sequential Monte Carlo – Move



- Resampling leaves us with duplicates
- Move each particle around a bit to diversify again
- In practice: run MCMC on the particles
 - Think again of MCMC kernels!

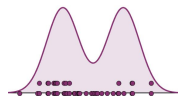
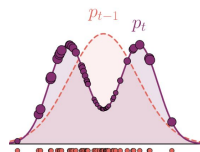
Sequential Monte Carlo – Pseudocode

Algorithm 1: General Sequential Monte Carlo sampler

Data: Number of particles N , sequence of T unnormalized densities $\gamma_1, \dots, \gamma_T$ with $p_t = \gamma_t / \mathcal{Z}_t$ with p_1 tractable and proper, and Markov kernels $\mathcal{K}_t$ with invariant density $p_t(\theta)$

Result: Samples $\{\theta_T^i\}_{i=1}^N$ approximating the target density $p_T(\theta)$, and evidence estimate $\hat{\mathcal{Z}}_T$

```
 $\hat{\mathcal{Z}}_1 \leftarrow 1;$   
for  $i \leftarrow 1$  to  $N$  do  
   $\theta_1^i \sim p_1(\theta);$                                 /* Generate initial samples */  
for  $t \leftarrow 2$  to  $T$  do  
  for  $i \leftarrow 1$  to  $N$  do  
     $w_t^i \leftarrow \gamma_t(\theta_{t-1}^i) / \gamma_{t-1}(\theta_{t-1}^i);$           /* Reweight */  
     $\hat{\mathcal{Z}}_t \leftarrow \hat{\mathcal{Z}}_{t-1} \times N^{-1} \sum_{i=1}^N w_t^i;$           /* Update evidence estimate */  
    for  $i \leftarrow 1$  to  $N$  do  
       $W_t^i \leftarrow w_t^i / \sum_{j=1}^N w_t^j;$                       /* Normalize weights */  
     $\{\tilde{\theta}_t^i\}_{i=1}^N \sim \text{resample}(\{\theta_{t-1}^i\}_{i=1}^N, \{W_t^i\}_{i=1}^N);$  /* Resample */  
    for  $i \leftarrow 1$  to  $N$  do  
       $\theta_t^i \leftarrow \mathcal{K}_t(\tilde{\theta}_t^i);$                             /* Move */
```



Sequential Monte Carlo – Paths

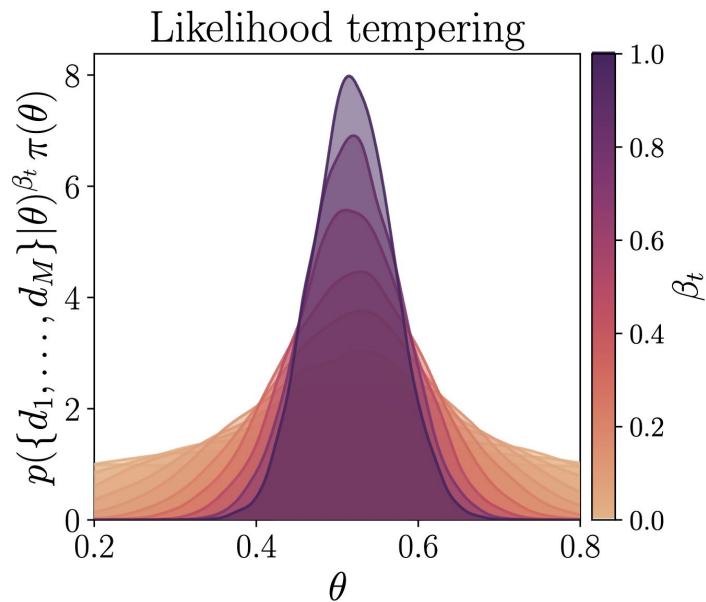
- So far we talked about “bridging” between “some” distributions
 - Sequential Monte Carlo **does not specify** this path $p_t, t=0,\dots,T$
- This leaves us with freedom!
- Two popular choices
 - Likelihood tempering
 - Data tempering
- Extra:
 - Model tempering (“posterior reuse”): see “Future” slides

Sequential Monte Carlo – Likelihood tempering

- Construct a path of “tempered” likelihood distributions (beta = inverse temperature)

$$p_t(\theta) = \frac{[\mathcal{L}(\theta)]^{\beta_t} \pi(\theta)}{\mathcal{Z}_t}$$

- Beta = 0 $\rightarrow$ prior
- Beta = 1 $\rightarrow$ posterior
- Gradually introduces the likelihood
- Weights: $w_t^i = \mathcal{L}(\theta)^{\beta_t - \beta_{t-1}}$

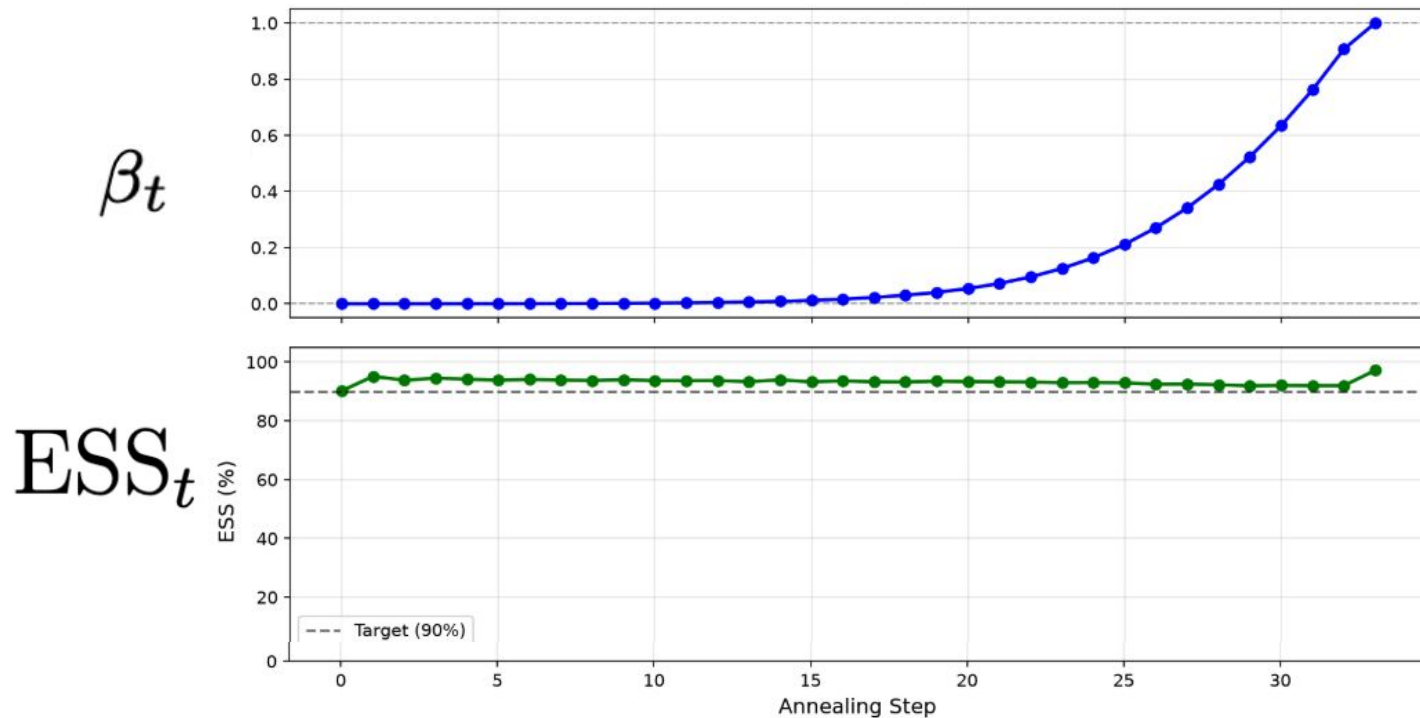


Sequential Monte Carlo – Likelihood tempering

- The temperature can be set **adaptively, guided by the ESS** $ESS_t = ESS(\{w_t^i\}_{i=1}^N)$
- Intuitively:
 - Larger beta jump $\rightarrow$ distributions more different $\rightarrow$ low ESS
 - Smaller beta jump $\rightarrow$ distributions similar $\rightarrow$ high ESS
- So: a user-defined target ESS: eg “90% of my N particles” alpha
 - Find the beta that gives this ESS: bisection search

$$\beta_t = \sup_{\beta_t \in [\beta_{t-1}, 1]} \{ESS_t \geq \alpha N\}$$

Sequential Monte Carlo – Likelihood tempering



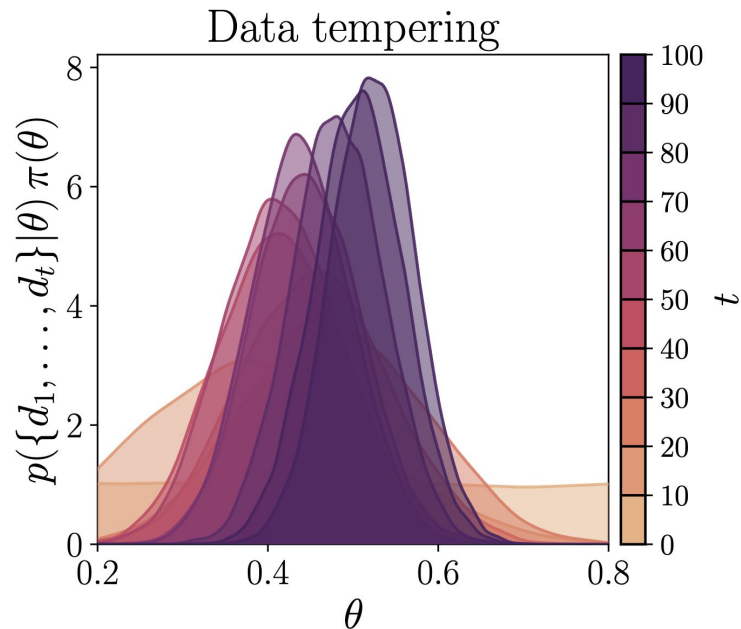
Sequential Monte Carlo – Data tempering

- Can be used when dataset “grows” $d_{a:b} = \{d_a, \dots, d_b\}$
 - E.g. lightcurves?
- Can traverse the *path of partial posteriors*

$$\mathcal{P}_t \equiv \frac{\mathcal{L}(d_{1:t}|\theta)\pi(\theta)}{\mathcal{Z}_t}$$

- Importance weight = single likelihood

$$w_t^i = \mathcal{L}_t(\theta_{t-1}^i)$$



Sequential Monte Carlo – Data tempering

- However, naive “add one datapoint at a time” is **inefficient**
 - Informative datapoint: ESS collapses
 - Non-informative datapoint: ESS remains very high
- So: generalize to **batches** of q datapoints

$$\mathcal{L}_{a:b}(\theta) = \prod_{i=a+1}^b \mathcal{L}_i(\theta)$$

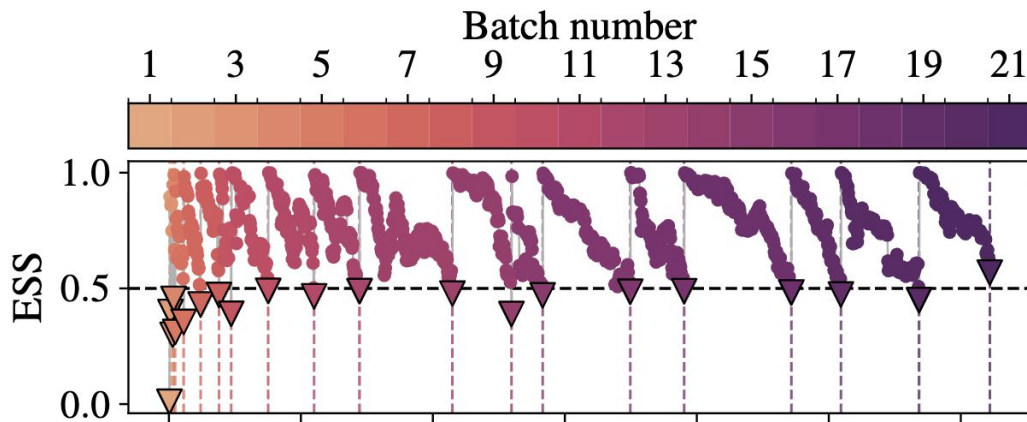
$$w_t^i(\theta_{t-1}^i) = \frac{\mathcal{L}_{1:b_t}(\theta_{t-1}^i)}{\mathcal{L}_{1:b_{t-1}}(\theta_{t-1}^i)} = \mathcal{L}_{b_{t-1}:b_t}(\theta_{t-1}^i)$$

Sequential Monte Carlo – Data tempering

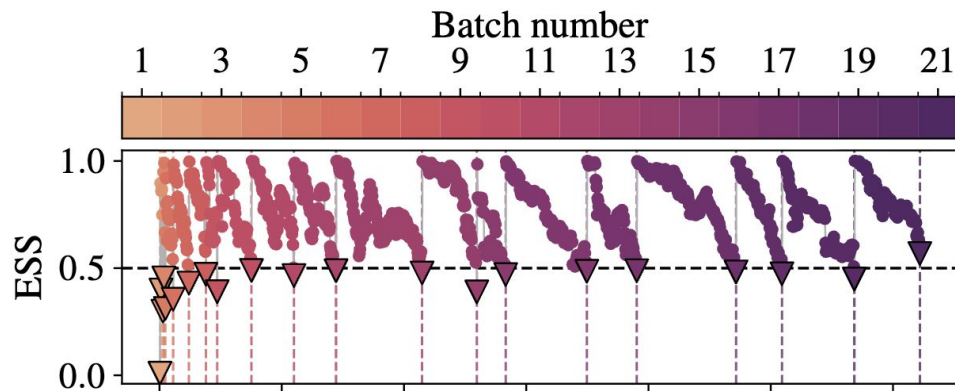
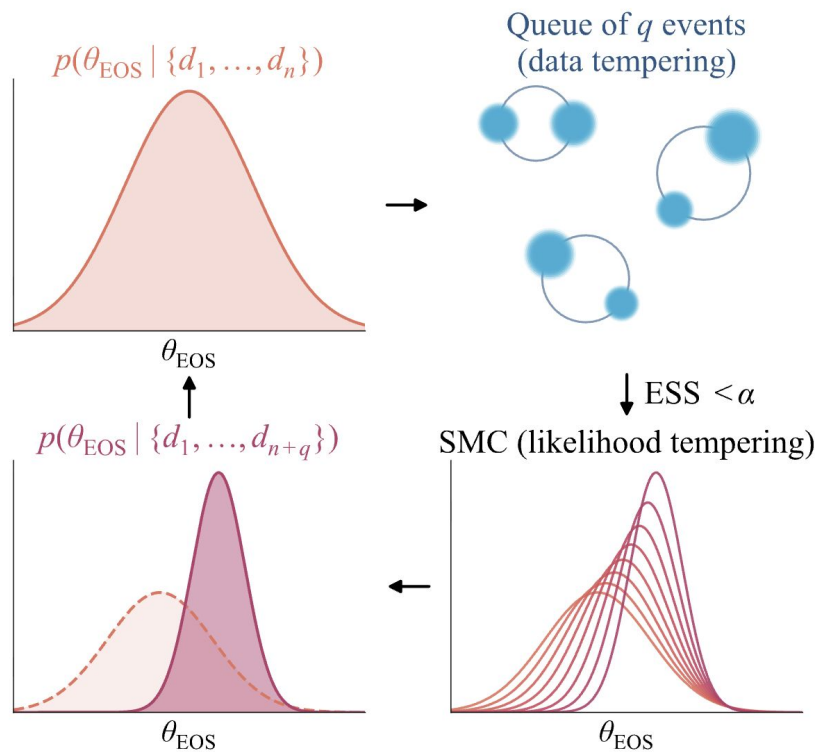
- Batch sizes can be set **adaptively, guided by the ESS**
- Update step each batch size: run SMC with likelihood tempering (LT)

$$b_t = \sup_{b_t \in [b_{t-1}, T]} \{\text{ESS}_t \geq \tilde{\alpha} N\}$$

$$\pi = \pi_0 \xrightarrow{\text{LT}} \mathcal{P}_{b_1} = \pi_{b_1} \xrightarrow{\text{LT}} \mathcal{P}_{b_2} = \pi_{b_2} \xrightarrow{\text{LT}} \dots \xrightarrow{\text{LT}} \mathcal{P}$$



Sequential Monte Carlo – Data tempering



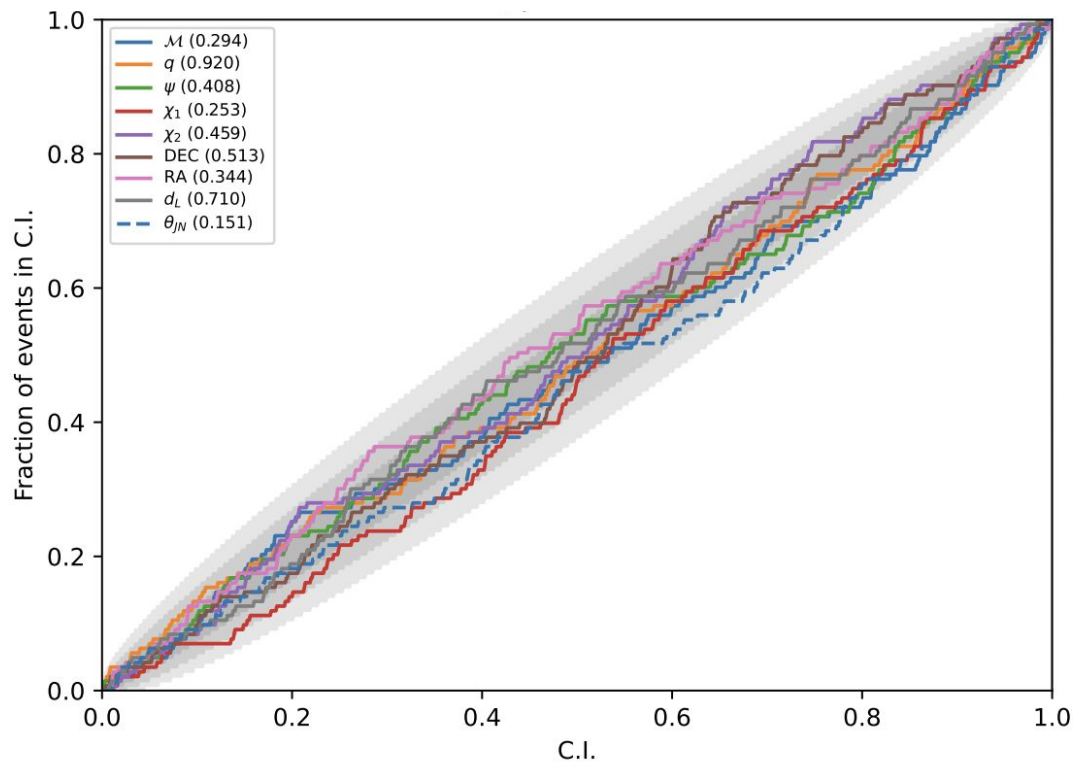
More...

- MCMC:
 - Hamiltonian MC
 - Normalizing flow
- Nested sampling:
 - Approximate likelihood-constrained prior
 - Dynamic nested sampling

Bayesian sharp bits

- Multiple inputs, multiple points of failure
 - Likelihood
 - Noise model
 - Maximum entropy distribution
 - Prior
 - Too informative
 - Sampling

Bayesian sharp bits

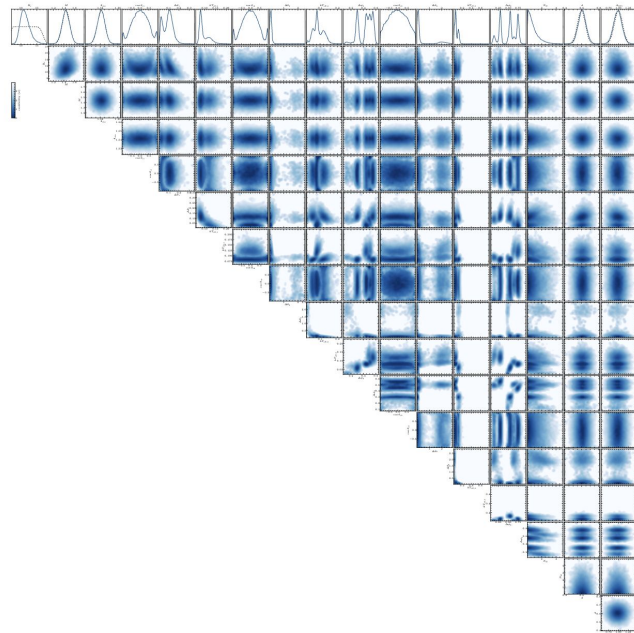


Computational methods

- Estimating the posterior distribution and/or compute Bayesian evidence
 - Monte Carlo integration and importance sampling
 - Markov chain Monte Carlo (MCMC)
 - Parallel-tempering MCMC
 - Nested sampling (→ NMMA today)
 - Sequential Monte Carlo
- **Future?**
 - **New samplers for NMMA**
 - ninjax

New samplers for NMMA?

- Bilby has “plugin” for samplers → wide array of samplers compatible
 - Eg pocoMC: now used in [GW](#) and even [NICER](#) (!)
- What about NMMA?
 - Can use this feature ***with 0 code changes***



New samplers for NMMA?

- What if pocoMC is actually far better at multimessenger posteriors than dynesty?

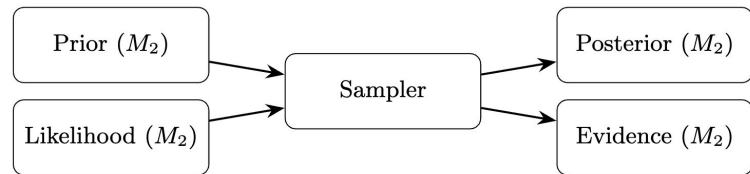


New samplers for NMMA?

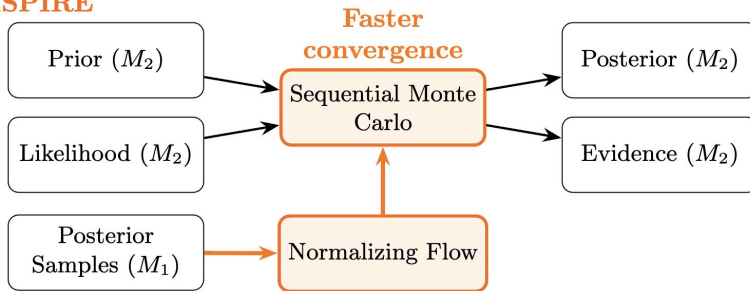
- SMC allows for “posterior reuse” ([ASPIRE](#)), compatible with bilby therefore NMMA
 - Start from previous posterior result
 - “Close” to new target posterior
- Sampling systematics for MMA inferences?
 - Many models (GW, KN, GRB, EOS...)!
 - → Many combinations

Starting point: Posterior Samples (M_1), New Model (M_2)

Traditional inference



ASPIRE



Computational methods

- Estimating the posterior distribution and/or compute Bayesian evidence
 - Monte Carlo integration and importance sampling
 - Markov chain Monte Carlo (MCMC)
 - Parallel-tempering MCMC
 - Nested sampling (→ NMMA today)
 - Sequential Monte Carlo
- **Future?**
 - New samplers for NMMA
 - **ninjax**

Future of NMMA – JAX-based?

- NMMA runs on CPU
- Can we port it GPUs, and make it super fast?
 - For GW? **YES! Jim:** <https://github.com/GW-JAX-Team/jim>
 - For EM? **YES! fiesta:** <https://github.com/nuclear-multimessenger-astronomy/fiestaEM>
 - For EOS? **YES! jester:** <https://github.com/nuclear-multimessenger-astronomy/jester>
 - For GW+EM+EOS? **NO!**
- **Need a single software stack combining these packages!**



Future of NMMA – JAX-based?

- Jim, fiesta, jester,... All based on JAX
 - Automatic differentiation: derivatives for free
 - Easy parallelization: “vmap”
 - CPU to GPU execution: with 0 code changes
- Python with “functional programming”
- Get GPU benefits “for free”
- Composable transformations
- Samplers: <https://github.com/blackjax-devs/blackjax>
 - Jim, fiesta, jester use it



JAX-based: does it work?



- Jim
 - Can run BNS PE in ~2 mins (vs ~10 hours CPU)
- Fiesta:
 - ~mins (surrogates)
- Jester
 - EOS inference ~minutes
 - Einstein Telescope projections feasible

Ninjax (NMMA in JAX)



Jim 



JESTER



Ninjax (NMMA in JAX)



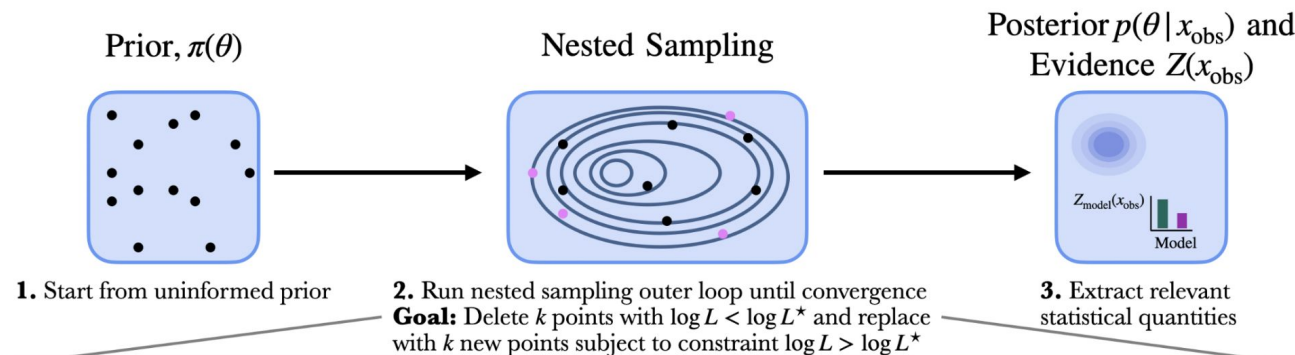
- Rough scaffolding: <https://github.com/nuclear-multimessenger-astronomy/ninjax>
- Need input from the group this week
 - How to integrate jim+fiesta+jester?
 - Roadmap for development?
- Easy to “combine” them?





Ninjax-SwiG: the ultimate MMA sampler?

- [paper](#)



A – Inner Kernel Development

$\mathcal{M}_c, q, \Lambda_1, \Lambda_2$
 s_1, s_2, t, t_c
 $d_L, \psi, \alpha, \delta$

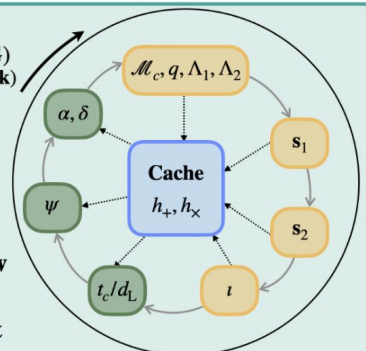
Custom Slice within Gibbs (SwiG)

- ✓ Improved mixing (This Work)
- ✓ Fast-slow optimisation
- ✓ Simple black-box integration
- ✗ Custom GW kernel

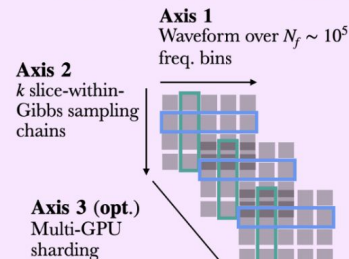
Standard Slice Sampling

- ✓ Simple
- ✓ Problem agnostic
- ✗ Poor mixing performance
- ✗ No structural optimisation

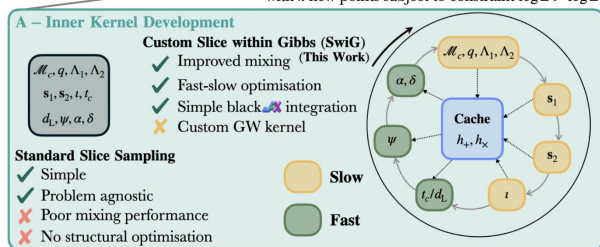
Slow
Fast



B – GPU Optimisation (This Work)



Ninjax-SwiG: why?



- Multimessenger inference has many “slow-fast” block structures
 - GW, EM, EOS:
 - GW slow
 - EM fast (surrogates)
 - GW detector calibration
 - Nuisance parameters eg systematic uncertainties Sahil
- Would this make high-dimensional scaling less problematic?
- What is the best blocking structure? Which params are heavily correlated?

Conclusions

NMMA: Bayesian inference on steroids

- Bayesian inference: posteriors and model selection on
 - GW, EM, EOS models
- Answering questions needs stochastic samplers
 - MCMC
 - Nested sampling
 - Sequential Monte Carlo
- GPU-based version of NMMA is within reach... within 1 week?
- Software x hardware can make leaps in scientific computing



Thanks for listening, and
keep moonfudging!

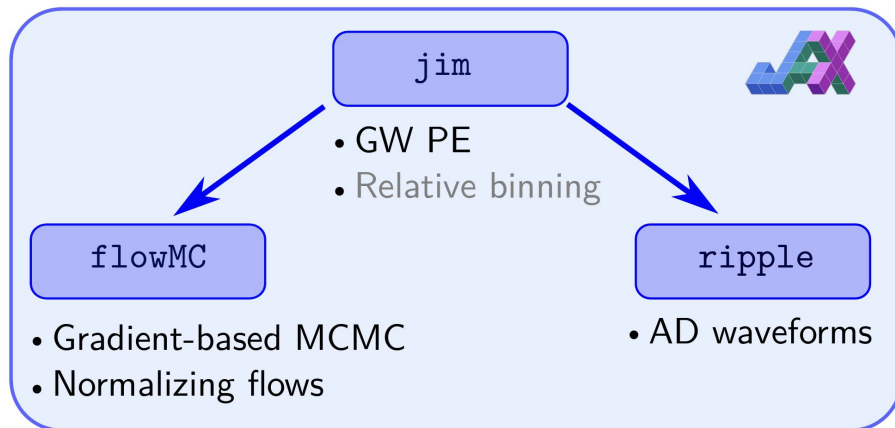
Normalizing flows

- Machine learning density estimators
- Samples $\rightarrow$ tractable densities
 - Generate new samples
 - Evaluate density at any point
- They are everywhere in Bayesian inference today!

Appendices

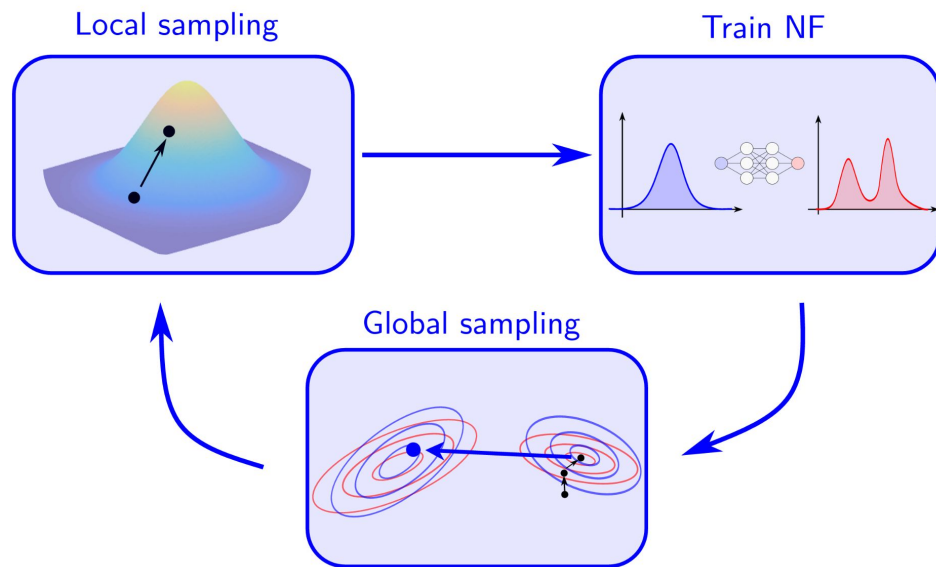
flowMC-jim-ripple approach

- Machine-learning enhanced sampling
- Auto-differentiable
- Highly portable with hardware acceleration on GPUs and TPUs



flowMC

- Local sampling
 - Traditional methods
 - Random walk
 - Gradient-based sampling
- Training the NF
 - The distribution of the walkers
- Global sampling
 - Proposal with the NF
- ??? -> Profit



Compare distributions

- How much information you gain from the experiment?
- Kullback–Leibler (KL) divergence:

$$D_{\text{KL}}(\mathcal{P}(\vec{\theta}) || \pi(\vec{\theta})) = \int d\vec{\theta} \mathcal{P}(\vec{\theta}) \log \left(\frac{\mathcal{P}(\vec{\theta})}{\pi(\vec{\theta})} \right)$$

- If the posterior is the same as the prior
 - KL divergence = 0
- If the posterior approaches a dirac delta, while the prior is not
 - KL divergence -> infinity
- In the unit of nats (natural log) and bits (base-2 log)

D_{KL}	Astro only	HIC only	Astro+HIC
$1.0n_{\text{sat}}$	0.079	0.109	0.270
$1.5n_{\text{sat}}$	0.075	0.108	0.266
$2.0n_{\text{sat}}$	0.112	0.019	0.174
$2.5n_{\text{sat}}$	0.244	0.006	0.274
$1.0M_{\odot}$	0.090	0.054	0.128
$1.4M_{\odot}$	0.185	0.022	0.210
$1.6M_{\odot}$	0.225	0.015	0.251
$2.0M_{\odot}$	0.228	0.008	0.222

Compare distributions

- Symmetrize the KL divergence
- Jensen–Shannon (JS) divergence:

$$\text{JSD}(P \parallel Q) = \frac{1}{2}D_{\text{KL}}(P \parallel M) + \frac{1}{2}D_{\text{KL}}(Q \parallel M),$$

$$\text{where } M = \frac{1}{2}(P + Q)$$

- If the two distributions are exactly the same
 - $\text{JSD} = 0$
- If the two distributions are completely different
 - $\text{JSD} = \log(2)$

Background-Foreground approach

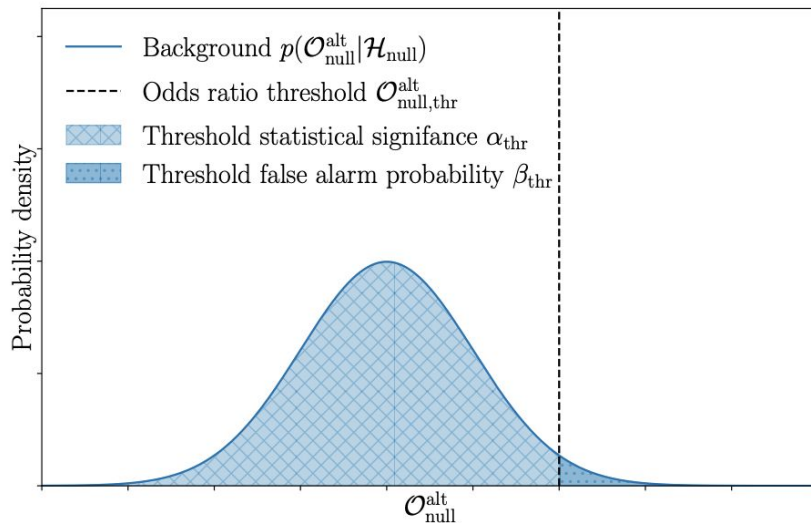
- Disadvantage of Bayesian hypothesis testing
 - Cannot calculate the actual probability
 - Fully trusting the noise model (likelihood function)
 - Hard to quantify uncertainty
- Frequentist approach:
 - Generate a distribution of testing statistic (Bayes factor) under the null hypothesis
 - Generate a distribution of testing statistic under the alternative hypothesis
 - Calculate the p-value

Background-Foreground approach

Threshold for claiming detection

$$\int_{\mathcal{O}}^{\infty} d\mathcal{O}_{\text{null}}^{\text{alt}} p(\mathcal{O}_{\text{null}}^{\text{alt}} | \mathcal{H}_{\text{null}}) = \beta = 1 - \alpha.$$

$$\int_{\mathcal{O}_{\text{null,thr}}^{\text{alt}}}^{\infty} d\mathcal{O}_{\text{null}}^{\text{alt}} p(\mathcal{O}_{\text{null}}^{\text{alt}} | \mathcal{H}_{\text{null}}) = \beta_{\text{thr}} = 1 - \alpha_{\text{thr}}.$$



Background-Foreground approach

Effectiveness of experiment

$$\xi = \int_{\mathcal{O}_{2,\text{thr}}^1}^{\infty} d\mathcal{O}_{\text{null}}^{\text{alt}} p(\mathcal{O}_{\text{null}}^{\text{alt}} | \mathcal{H}_2).$$

